

Department of Information Engineering
MSc in Computer Engineering (a.a. 2022/2023)
Computer Architecture (Prof. Cosimo Antonio Prete)

Introduction To Quantum Computing

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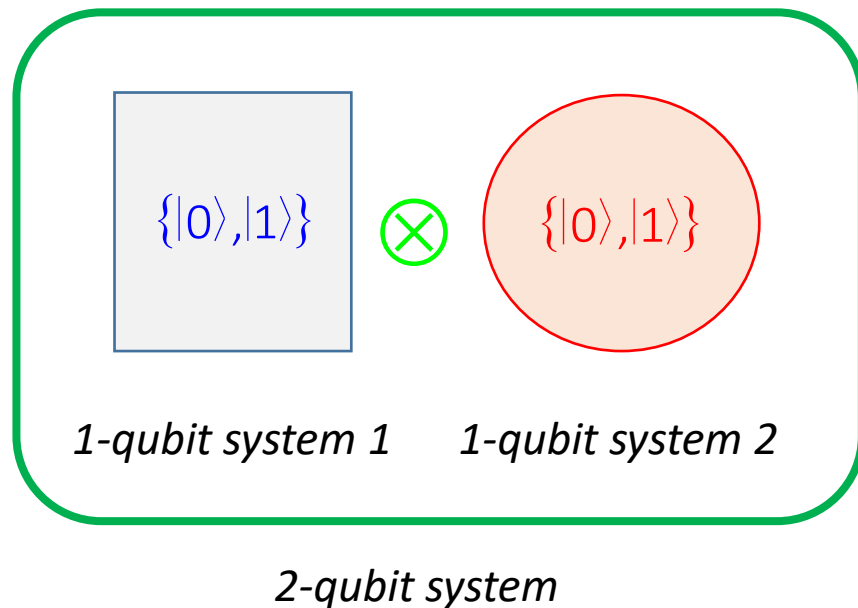


Recap From Previous Seminar

Composite Systems

Postulate 4:

- The state space of a *composite physical system* is the *tensor product* of the state spaces of the *component physical systems*.

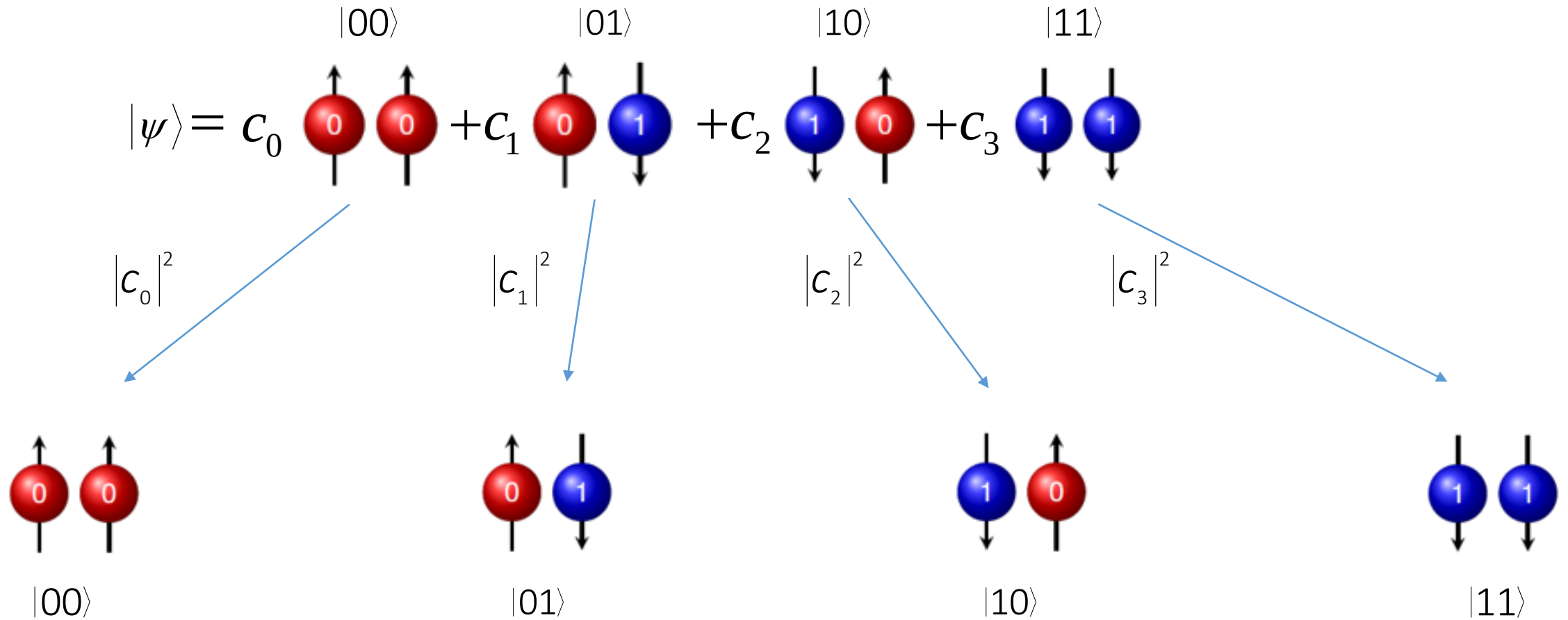


Moreover, if we have systems numbered 1 through n , and system number i is prepared in the state $|\psi_i\rangle$, then the joint state of the total system is $|\psi_1\rangle \otimes |\psi_2\rangle \otimes \cdots \otimes |\psi_n\rangle$



Thus, the tensor product is the mathematical structure used to describe the state space of a composite physical system

Multiple Qubits Measurement



$$|c_0|^2 + |c_1|^2 + |c_2|^2 + |c_3|^2 = 1$$

Measuring Individual Qubits

- For a **two** qubit system, we could measure, for example, the first qubit

$$|\psi\rangle = c_0|00\rangle + c_1|01\rangle + c_2|10\rangle + c_3|11\rangle$$

- Measuring the **first** qubit alone gives 0 with probability

$$|c_0|^2 + |c_1|^2$$

leaving the post-measurement state

$$|\psi\rangle = \frac{c_0|00\rangle + c_1|01\rangle}{\sqrt{|c_0|^2 + |c_1|^2}}$$

- Note how the post-measurement state is **re-normalized**

Entanglement

- There exist quantum states that *cannot be factored into product states*
- These are called *entangled states*
- The *entanglement* is one of the most interesting and puzzling ideas associated with composite quantum systems
- Consider the two qubit state named the *Bell State* or *EPR pair*

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \rightarrow \frac{1}{\sqrt{2}} \begin{array}{c} \uparrow \\ \text{0} \end{array} \begin{array}{c} \uparrow \\ \text{0} \end{array} + \frac{1}{\sqrt{2}} \begin{array}{c} \downarrow \\ \text{1} \end{array} \begin{array}{c} \downarrow \\ \text{1} \end{array}$$

Two-Qubit Quantum Gates

- Quantum gates can also operate on two qubits at the same time
- Some important examples include:
 1. The *CNOT gate* or *controlled-NOT gate*, or since the X gate is the *NOT* gate, the *CNOT gate* is also called the *cX gate* or *controlled-X gate*
 2. The controlled- U gate
 3. The *SWAP* gate

Multi-Gate Circuits

- We have all the ingredients to make countless quantum circuits from the basic binary gates introduced above
- We'll start with the famous Bell states
- These are the only four **maximally** entangled pairs, and they are known as the *Bell states* or *EPR states* or *EPR pairs* (for the physicists Einstein, Podolsky, and Rosen)

A Circuit that Produces Bell States

- We've already met them

$$|\beta_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \rightarrow |\Phi^+\rangle$$

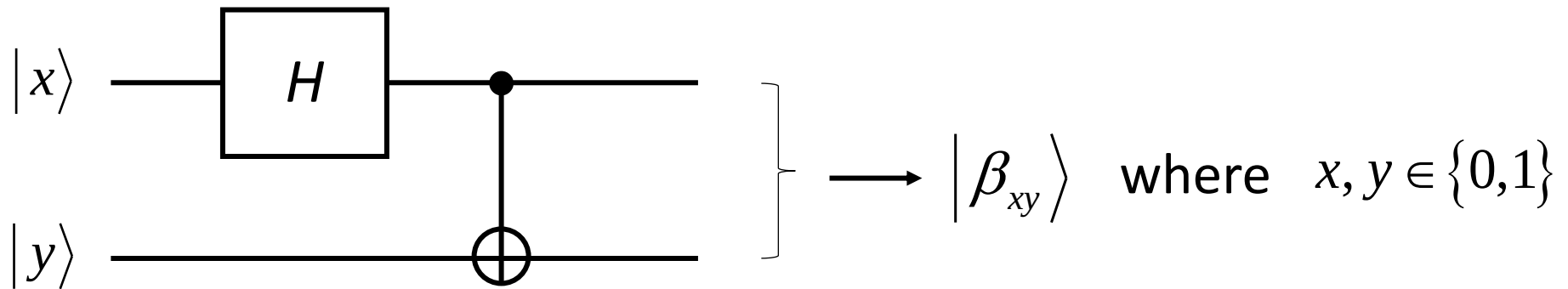
$$|\beta_{01}\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \rightarrow |\Psi^+\rangle$$

$$|\beta_{10}\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \rightarrow |\Phi^-\rangle$$

$$|\beta_{11}\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \rightarrow |\Psi^-\rangle$$

A Circuit that Produces Bell States

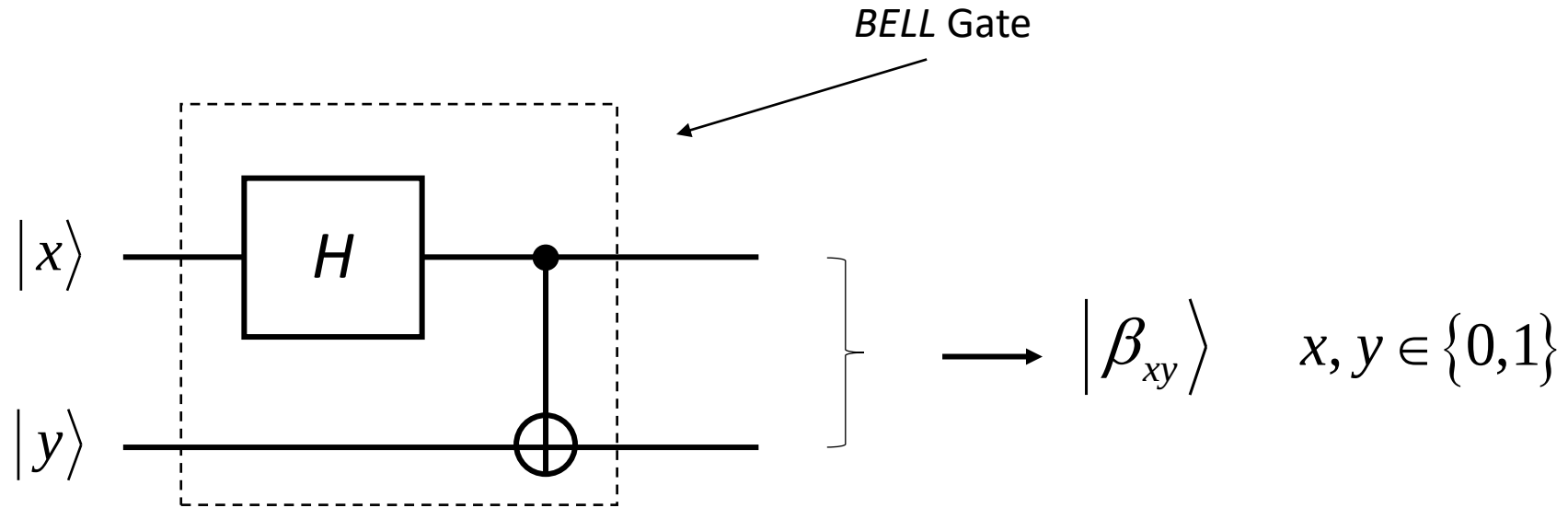
- The circuit that produces these four states, using the standard CBS basis as inputs $\{|0\rangle|0\rangle, |0\rangle|1\rangle, |1\rangle|0\rangle, |1\rangle|1\rangle\}$ is



which can be seen as a combination of a unary Hadamard gate with a CNOT gate

A Circuit that Produces Bell States

- We could emphasize that this is a binary gate in its own right by calling it BELL and boxing it



A Circuit that Produces Bell States

- In concrete terms, the algebraic expression,

$$BELL(|x\rangle|y\rangle) = |\beta_{xy}\rangle$$

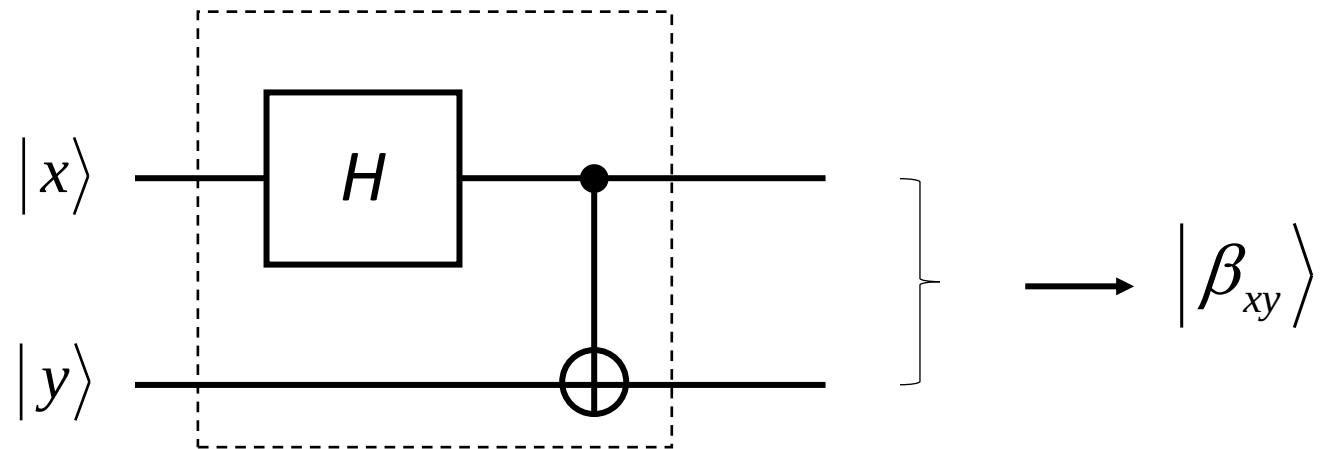
is telling us that

$$BELL(|0\rangle|0\rangle) = |\beta_{00}\rangle$$

$$BELL(|0\rangle|1\rangle) = |\beta_{01}\rangle$$

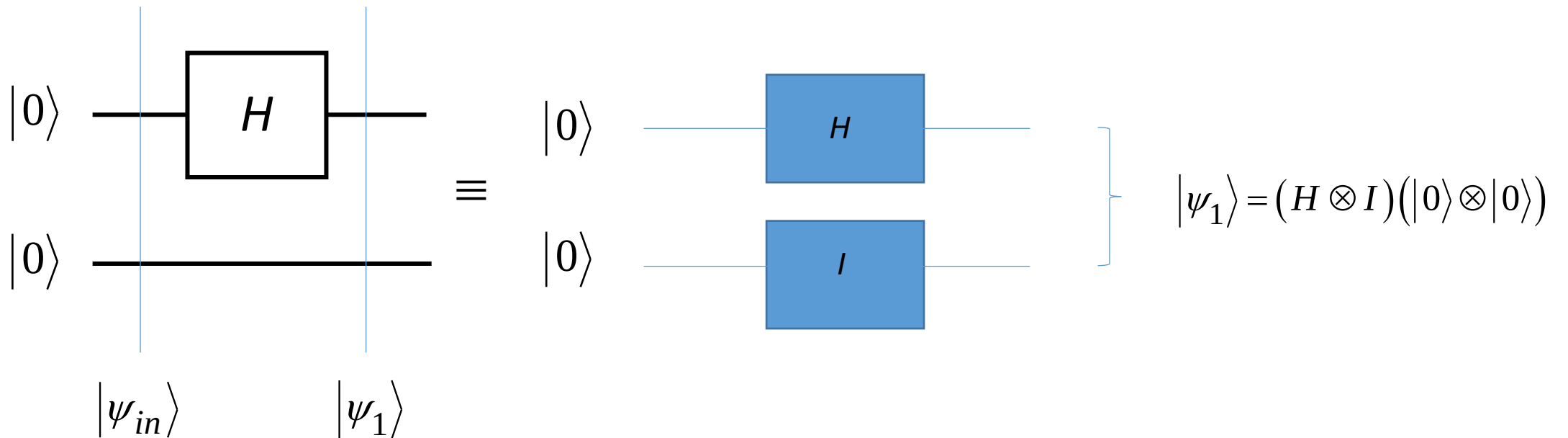
$$BELL(|1\rangle|0\rangle) = |\beta_{10}\rangle$$

$$BELL(|1\rangle|1\rangle) = |\beta_{11}\rangle$$



A Circuit that Produces Bell States

- Let's prove that

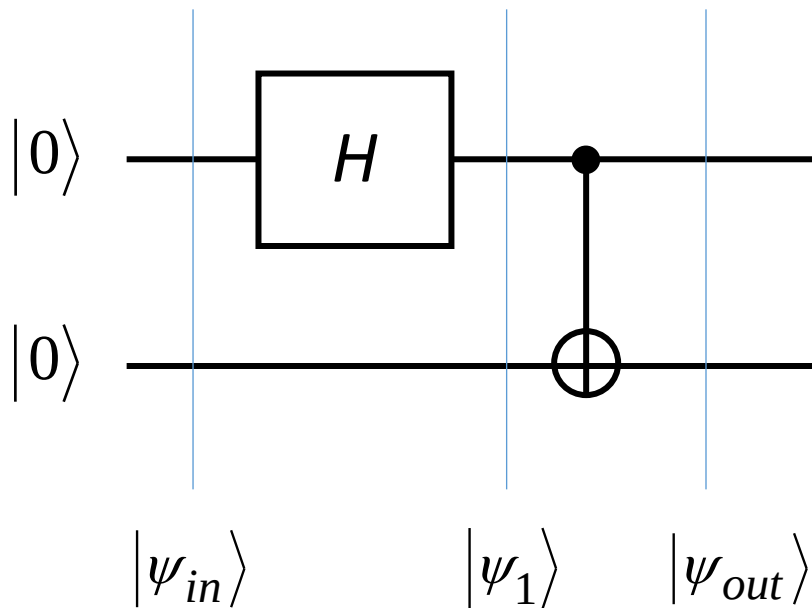


A Circuit that Produces Bell States

$$H|x\rangle = \frac{|0\rangle + (-1)^x |1\rangle}{\sqrt{2}}$$

$$\forall x \in \{0,1\}$$

- Let's prove that $BELL(|0\rangle|0\rangle) = |\beta_{00}\rangle$



$$|\psi_{in}\rangle = |0\rangle|0\rangle$$

$$|\psi_1\rangle = (H \otimes I)(|0\rangle \otimes |0\rangle) = H|0\rangle \otimes I|0\rangle$$

$$= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$$

$$|\psi_{out}\rangle = CNOT|\psi_1\rangle = \frac{1}{\sqrt{2}}(CNOT|00\rangle + CNOT|10\rangle)$$

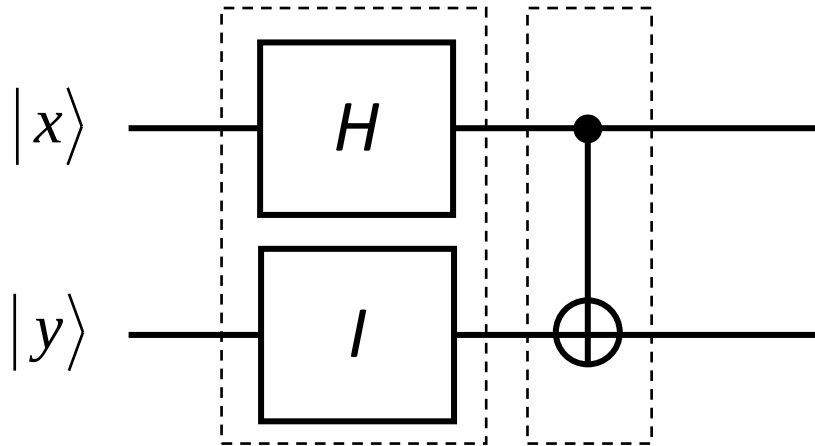
$$= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \equiv \beta_{00}$$

The Matrix for BELL

- The matrix for this gate can be constructed using the various techniques we have already studied
- For example, you can use the standard linear algebra approach of building columns for the matrix from the four outputs of the gate
- For variety, let's take a different path
- The Hadamard gate has a plain quantum wire below it, meaning that such a quantum wire is implicitly performing an identity operator at that point

The Matrix for BELL

- So we could write the gate using the equivalent symbolism



Explain why the order of the matrices is opposite the order of the gates in the diagram

$$BELL = (CNOT)(H \otimes I) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{bmatrix}$$

The Matrix for BELL

- Let's apply the BELL matrix to the input state $|10\rangle$

$$BELL|10\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle) = \beta_{10}$$

- In general, it can be shown that the *BELL* matrix gives the four Bell states when one presents the four CBS states as inputs

The Matrix for BELL Dagger

- It can also be proved easily that the *BELL* matrix is unitary $BELL^\dagger$
- Using the adjoint conversion rules, and remembering that the order of operators in the circuit is opposite of that in the algebra, we find

$$BELL^\dagger = [(CNOT)(H \otimes I)]^\dagger = (H \otimes I)^\dagger (CNOT)^\dagger = (H \otimes I)(CNOT)$$

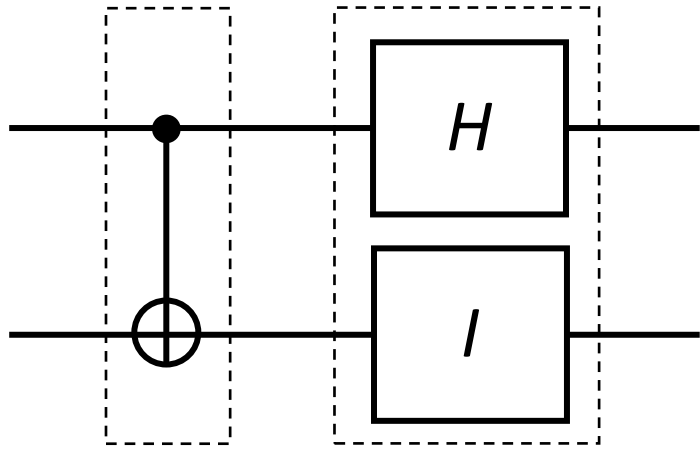
- The final equality is a consequence of the fact that CNOT and $H \otimes I$ are both self-adjoint and that

$$(H \otimes I)^\dagger = H^\dagger \otimes I^\dagger = H \otimes I$$

$$(CNOT)^\dagger = CNOT$$

The Matrix for BELL Dagger

- In other words, we just reverse the order of the two sub-operators that comprise BELL
- This makes the circuit diagram for $BELL^\dagger$ come out to be



$$BELL^\dagger = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{bmatrix}^\dagger = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 \end{bmatrix}$$

Four Bell States from One

- We can now list the following results

$$(I \otimes I)|\beta_{00}\rangle = |\beta_{00}\rangle$$

$$(X \otimes I)|\beta_{00}\rangle = |\beta_{01}\rangle$$

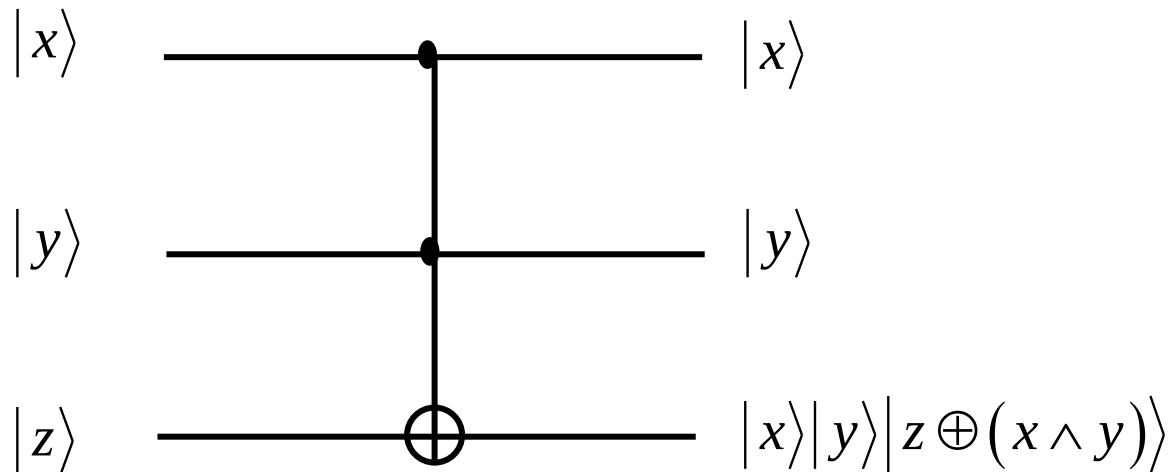
$$(Z \otimes I)|\beta_{00}\rangle = |\beta_{10}\rangle$$

$$(iY \otimes I)|\beta_{00}\rangle = |\beta_{11}\rangle$$

- Thus a local operation on an entangled state changes the entire state, affecting both qubits of the entangled pair

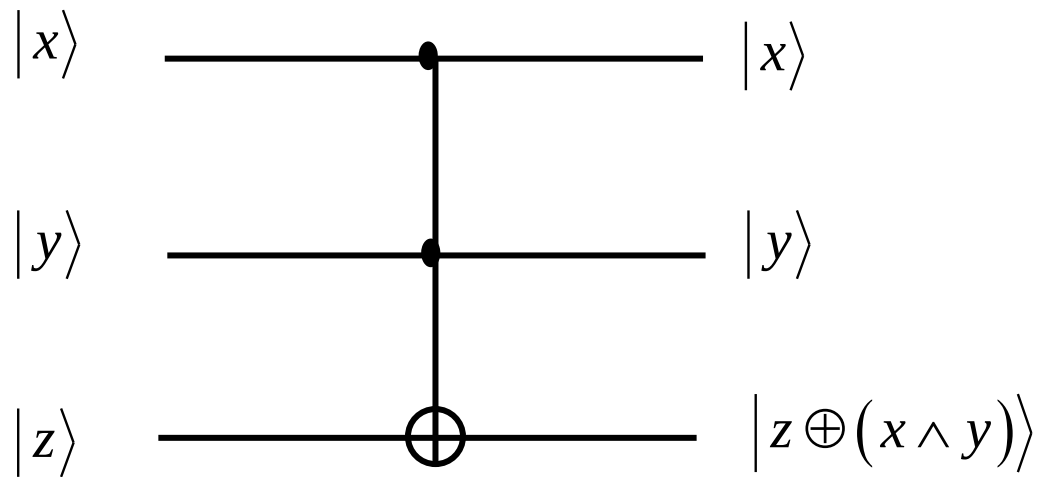
Toffoli Gate

- This is similar to the CNOT gate, but with two controlling bits
- The bottom bit flips only when *both* of the top two bits are in state $|1\rangle$
- We can write this operation as taking state $|x\rangle|y\rangle|z\rangle$ to $|x\rangle|y\rangle|z \oplus (x \wedge y)\rangle$



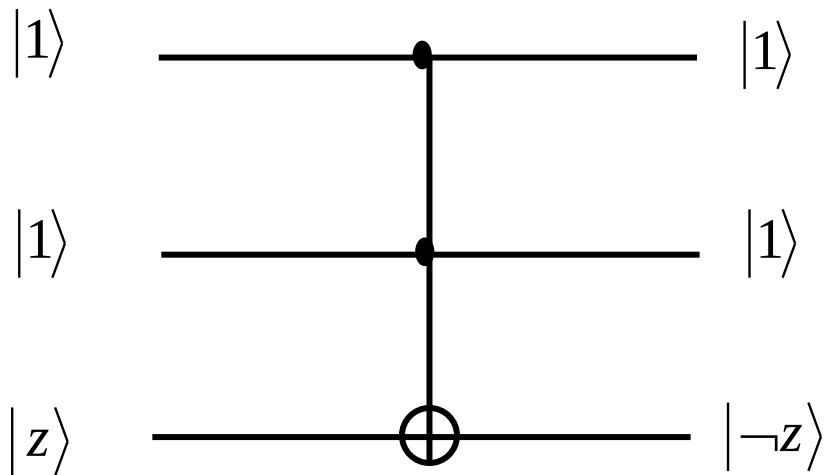
Toffoli Gate

- The *AND* gate is obtained by setting the bottom $|z\rangle$ input to $|0\rangle$



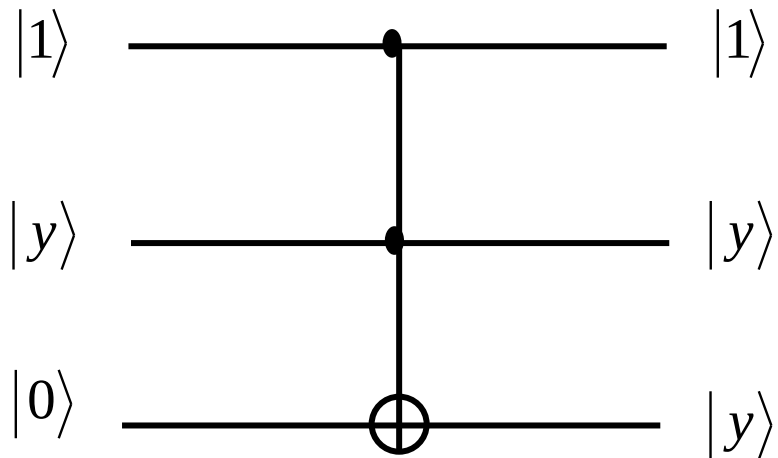
Toffoli Gate

- The *NOT* gate is obtained by setting the top two inputs to $|1\rangle$
- The bottom output will be $|(1 \wedge 1) \oplus z\rangle = |1 \oplus z\rangle = |\neg z\rangle$



Toffoli Gate

- In order to construct all gates, we must also have a way of producing a fanout of values.
- In other words, a gate is needed that inputs a value and outputs two of the same values.
- This can be obtained by the following Toffoli setting



Toffoli Gate

- The Toffoli gate flips the right qubit if the left and middle qubits are 1

$$\text{Toffoli}|000\rangle = |000\rangle$$

$$\text{Toffoli}|001\rangle = |001\rangle$$

$$\text{Toffoli}|010\rangle = |010\rangle$$

$$\text{Toffoli}|011\rangle = |011\rangle$$

$$\text{Toffoli}|100\rangle = |100\rangle$$

$$\text{Toffoli}|101\rangle = |101\rangle$$

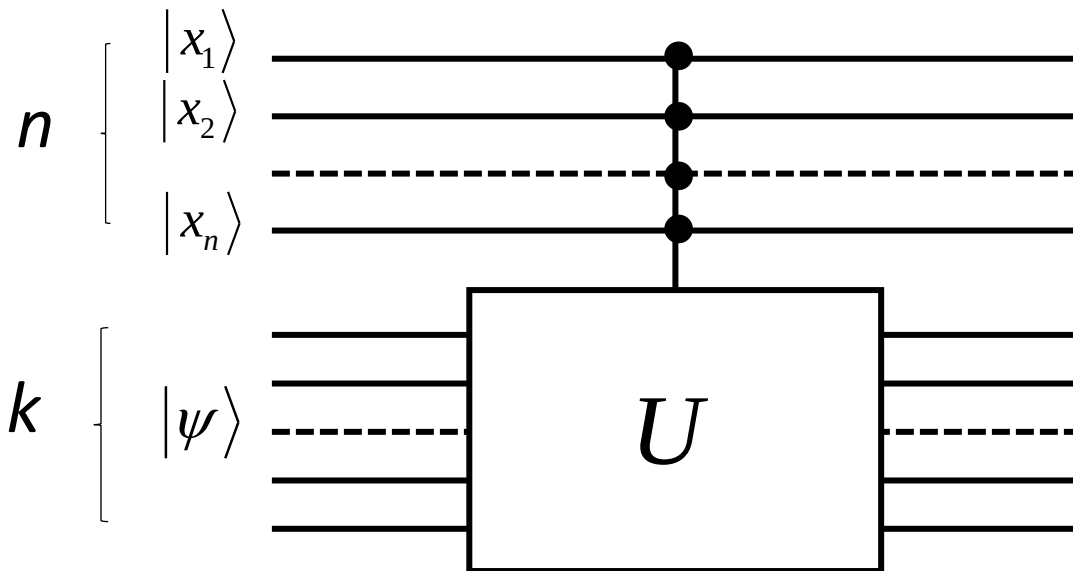
$$\text{Toffoli}|110\rangle = |111\rangle$$

$$\text{Toffoli}|111\rangle = |110\rangle$$

	000	001	010	011	100	101	110	111
000	1	0	0	0	0	0	0	0
001	0	1	0	0	0	0	0	0
010	0	0	1	0	0	0	0	0
011	0	0	0	1	0	0	0	0
100	0	0	0	0	1	0	0	0
101	0	0	0	0	0	0	1	0
110	0	0	0	0	0	1	0	0
111	0	0	0	0	0	0	0	1

Conditioning On Multiple Qubits

- Now that we know how to condition on a single qubit being set, what about conditioning on multiple qubits?
- More generally, suppose we have $n + k$ qubits, and U is a k qubit unitary operator



Then we define the controlled operation $C^n(U)$ by the equation

$$C^n(U) |x_1 x_2 x_3 \cdots x_n\rangle |\psi\rangle = |x_1 x_2 x_3 \cdots x_n\rangle U^{x_1 x_2 x_3 \cdots x_n} |\psi\rangle$$

Universal Quantum Gates

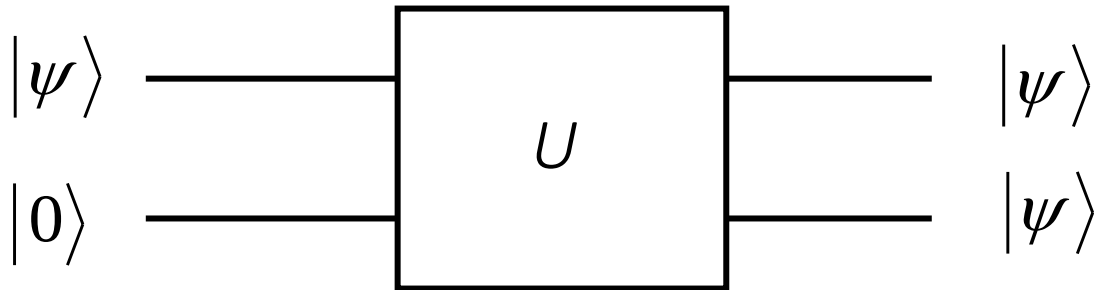
- There are many interesting quantum gates other than the CNOT
- However, in a sense the CNOT and single qubit gates are the prototypes for *all* other gates because of the following remarkable *universality* result
- *Any multiple qubit logic gate may be composed from **CNOT** and **single qubit gates***
- The proof is omitted

Universal Quantum Gates

- A *universal gate set* allows us to approximate any *unitary* to any desired precision
- Proving that a set of gates is universal is an advanced subject, which extends beyond the scope of our seven seminars
- Four examples of universal gate sets are
 - $\{R_x, R_y, R_z, Ph, CNOT\}$
 - $\{H, S, T, CNOT\}$
 - $\{R_{\pi/8}, S, CNOT\}$
 - $\{Toffoli, H, S\}$

No-Cloning Theorem

- Is it possible to make a copy of an *unknown* quantum state?
- Surprisingly, it turns out that the answer to this question is *NO*
- Suppose we try to copy a qubit in the *unknown* state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$



or

$$U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$$

No-Cloning Theorem

- By using linear algebra

$$\begin{bmatrix} U_{00} & U_{01} & U_{02} & U_{03} \\ U_{10} & U_{11} & U_{12} & U_{13} \\ U_{20} & U_{21} & U_{22} & U_{23} \\ U_{30} & U_{31} & U_{32} & U_{33} \end{bmatrix} \left(\begin{bmatrix} \alpha \\ \beta \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \otimes \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

- Since

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \alpha \\ 0 \\ \beta \\ 0 \end{bmatrix}, \quad \text{and} \quad \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \otimes \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha^2 \\ \alpha\beta \\ \beta\alpha \\ \beta^2 \end{bmatrix} \quad \Rightarrow$$

No-Cloning Theorem

$$\begin{bmatrix} U_{00} & U_{01} & U_{02} & U_{03} \\ U_{10} & U_{11} & U_{12} & U_{13} \\ U_{20} & U_{21} & U_{22} & U_{23} \\ U_{30} & U_{31} & U_{32} & U_{33} \end{bmatrix} \begin{bmatrix} \alpha \\ 0 \\ \beta \\ 0 \end{bmatrix} = \begin{bmatrix} \alpha^2 \\ \alpha\beta \\ \alpha\beta \\ \beta^2 \end{bmatrix} \quad \longrightarrow \quad \begin{bmatrix} U_{00}\alpha + U_{02}\beta \\ U_{10}\alpha + U_{12}\beta \\ U_{20}\alpha + U_{22}\beta \\ U_{30}\alpha + U_{32}\beta \end{bmatrix} = \begin{bmatrix} \alpha^2 \\ \alpha\beta \\ \alpha\beta \\ \beta^2 \end{bmatrix}$$

There are many possible solutions, such as

$$\begin{aligned} U_{00} &= \alpha, & U_{02} &= 0, & U_{10} &= 0, & U_{12} &= \alpha \\ U_{20} &= 0, & U_{22} &= \alpha, & U_{30} &= 0, & U_{32} &= \beta \end{aligned}$$

but this requires knowing α and β ,
which we do not know

Any general solution requires α
and β , so there is no operator U
that allows us to copy a
general, *unknown* quantum
state

No-Cloning Theorem

$$\begin{aligned} U_{00} &= \alpha, & U_{02} &= 0, & U_{10} &= 0, & U_{12} &= \alpha \\ U_{20} &= 0, & U_{22} &= \alpha, & U_{30} &= 0, & U_{32} &= \beta \end{aligned}$$

- Based on the results obtained previously the U matrix is

$$U = \begin{bmatrix} \alpha & U_{01} & 0 & U_{03} \\ 0 & U_{11} & \alpha & U_{13} \\ 0 & U_{21} & \alpha & U_{23} \\ 0 & U_{31} & \beta & U_{33} \end{bmatrix} \quad U_{ij} \neq \{0, \alpha, \beta\}$$

- Now assume that *we know* the state that we want to clone, e.g.

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad \longrightarrow \quad \alpha = \beta = \frac{1}{\sqrt{2}}$$

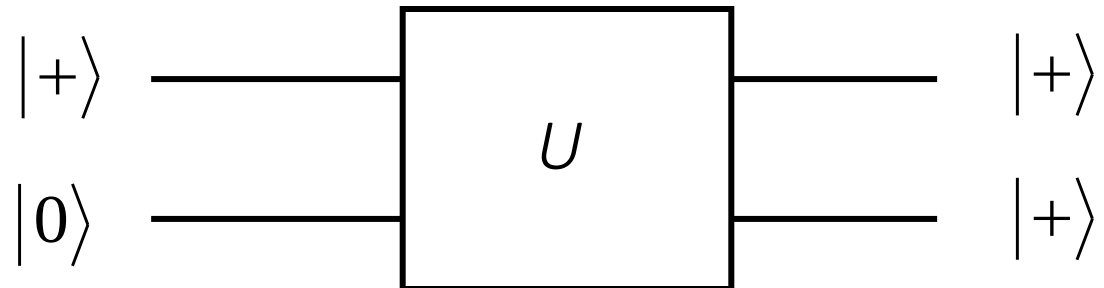
No-Cloning Theorem

- Therefore

$$U = \begin{bmatrix} 1/\sqrt{2} & U_{01} & 0 & U_{03} \\ 0 & U_{11} & 1/\sqrt{2} & U_{13} \\ 0 & U_{21} & 1/\sqrt{2} & U_{23} \\ 0 & U_{31} & 1/\sqrt{2} & U_{33} \end{bmatrix}$$

- Let's verify that

$$U(|+\rangle|0\rangle) = |+\rangle|+\rangle \quad \longrightarrow$$



No-Cloning Theorem

Since

$$|+\rangle|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad \longrightarrow$$
$$U(|+\rangle|0\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{2} & U_{01} & 0 & U_{03} \\ 0 & U_{11} & 1/\sqrt{2} & U_{13} \\ 0 & U_{21} & 1/\sqrt{2} & U_{23} \\ 0 & U_{31} & 1/\sqrt{2} & U_{33} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = |+\rangle|+\rangle$$

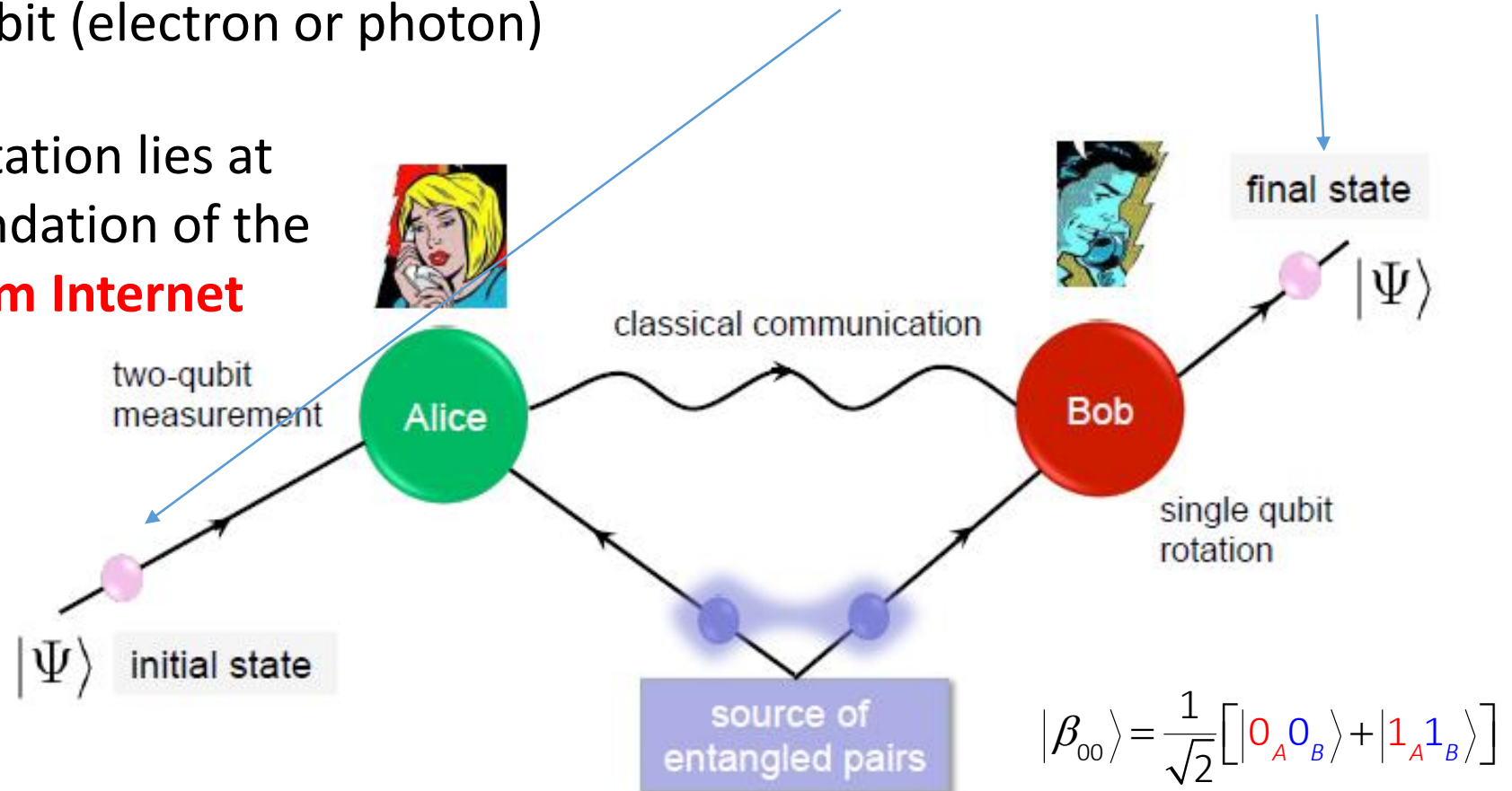
Thus, we can produce additional copies of $|+\rangle$

Quantum Teleportation

Quantum Teleportation

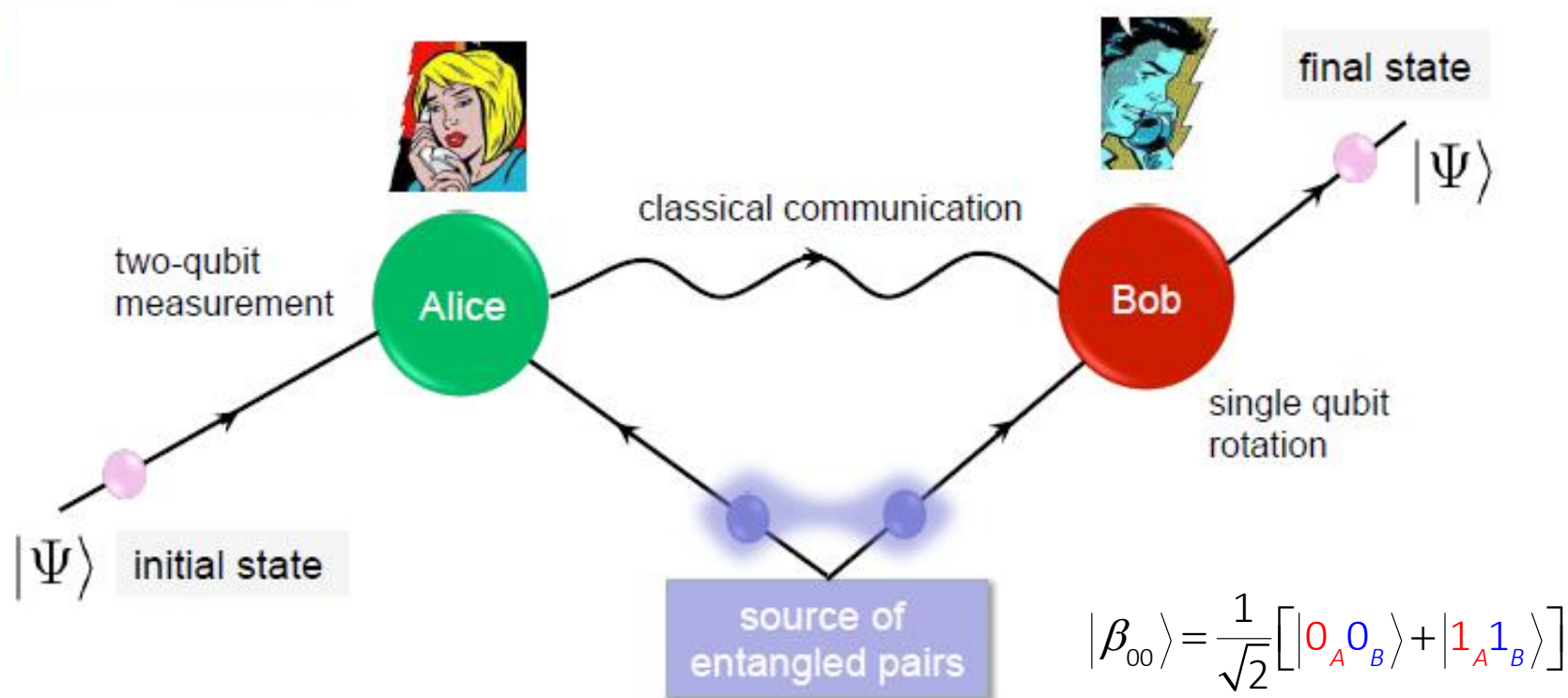
- Teleportation refers to an operation in which the quantum **state** of a qubit (confined single electron or photon) dissolves **here** and reappears **there**, on a different qubit (electron or photon)

Teleportation lies at the foundation of the **Quantum Internet**

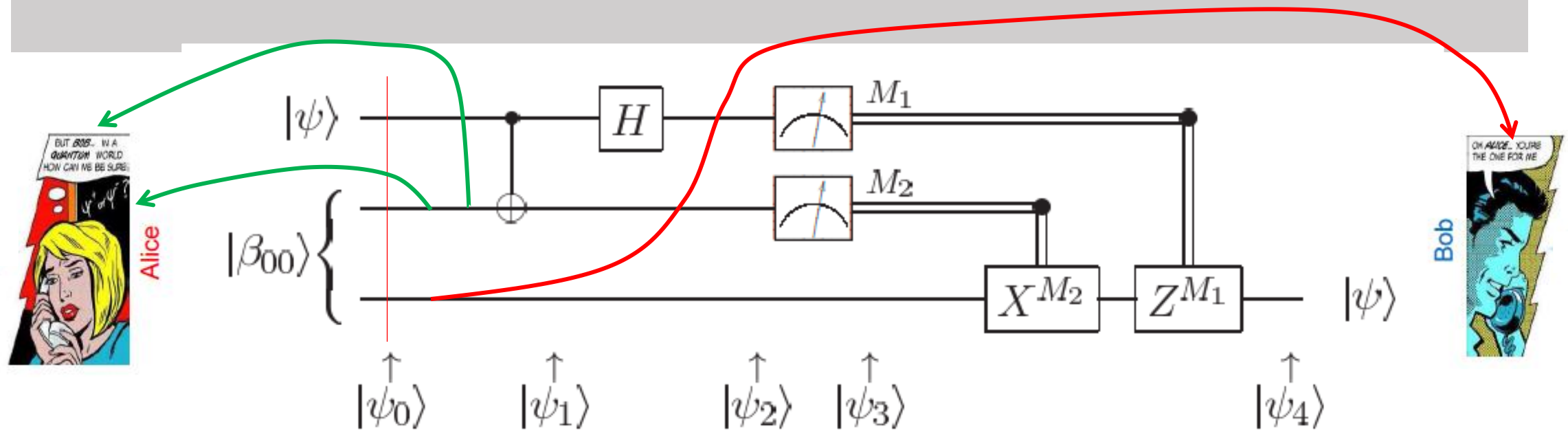


Quantum Teleportation

- Only the quantum *state* moves; the electron or other physical qubit remains where it was, and the receiver can in fact be a very different form of physical qubit than the sender



Quantum Teleportation

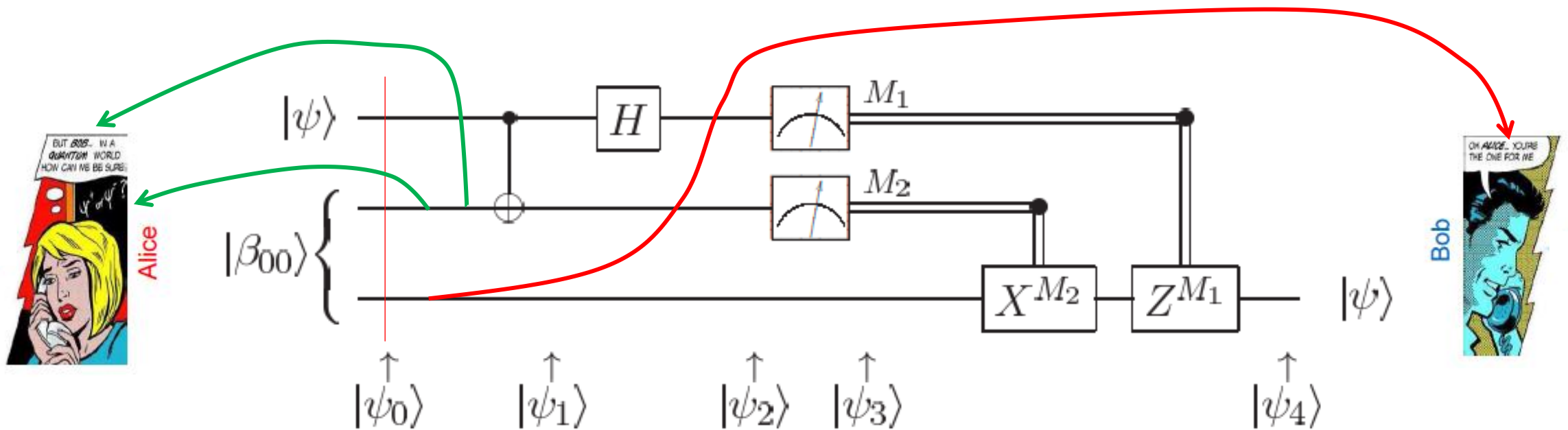


- The state to be teleported is $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where α and β are unknown amplitudes
- The state input into the circuit

$$|\psi_0\rangle = |\psi\rangle|\beta_{00}\rangle = (\alpha|0\rangle + \beta|1\rangle) \left(\frac{|0_A 0_B\rangle + |1_A 1_B\rangle}{\sqrt{2}} \right)$$

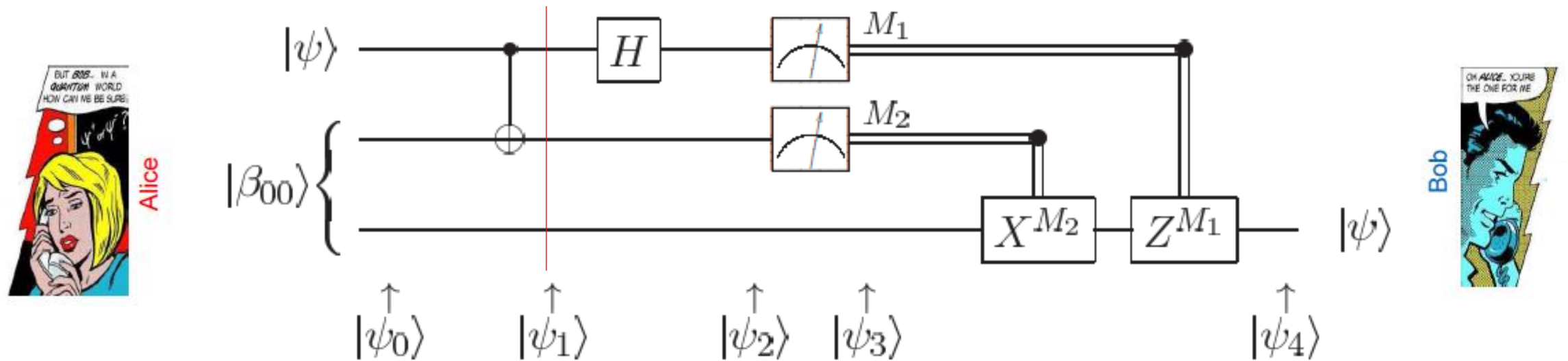
Quantum Teleportation

$$|\psi_0\rangle = |\psi\rangle |\beta_{00}\rangle = (\alpha|0\rangle + \beta|1\rangle) \left(\frac{|0_A 0_B\rangle + |1_A 1_B\rangle}{\sqrt{2}} \right)$$



Thus
$$|\psi_0\rangle = \frac{1}{\sqrt{2}} \left[\alpha |0_A\rangle (|0_A 0_B\rangle + |1_A 1_B\rangle) + \beta |1_A\rangle (|0_A 0_B\rangle + |1_A 1_B\rangle) \right]$$

Quantum Teleportation



- Alice sends her qubits through a CNOT gate, obtaining

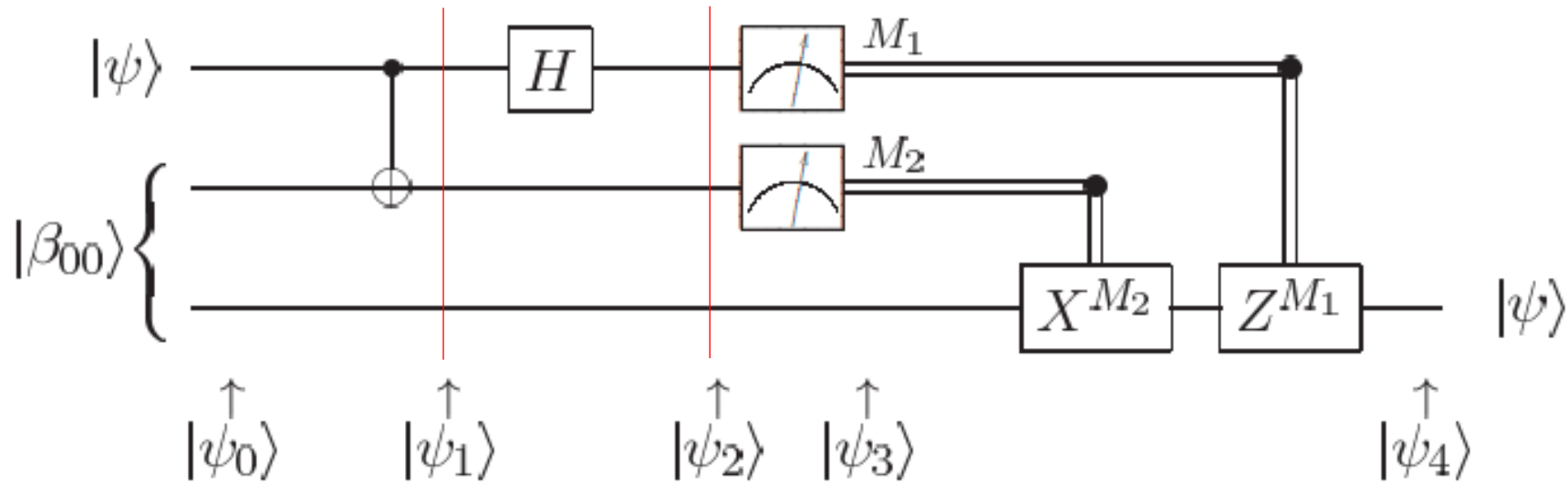
$$|\psi_0\rangle = \frac{1}{\sqrt{2}} \left[\alpha |0_A\rangle (|0_A 0_B\rangle + |1_A 1_B\rangle) + \beta |1_A\rangle (|0_A 0_B\rangle + |1_A 1_B\rangle) \right]$$

$$|\psi_1\rangle = \frac{1}{\sqrt{2}} \left[\alpha |0_A\rangle (|0_A 0_B\rangle + |1_A 1_B\rangle) + \beta |1_A\rangle (|1_A 0_B\rangle + |0_A 1_B\rangle) \right]$$

CNOT ↓ ↓ CNOT



Alice



Bob

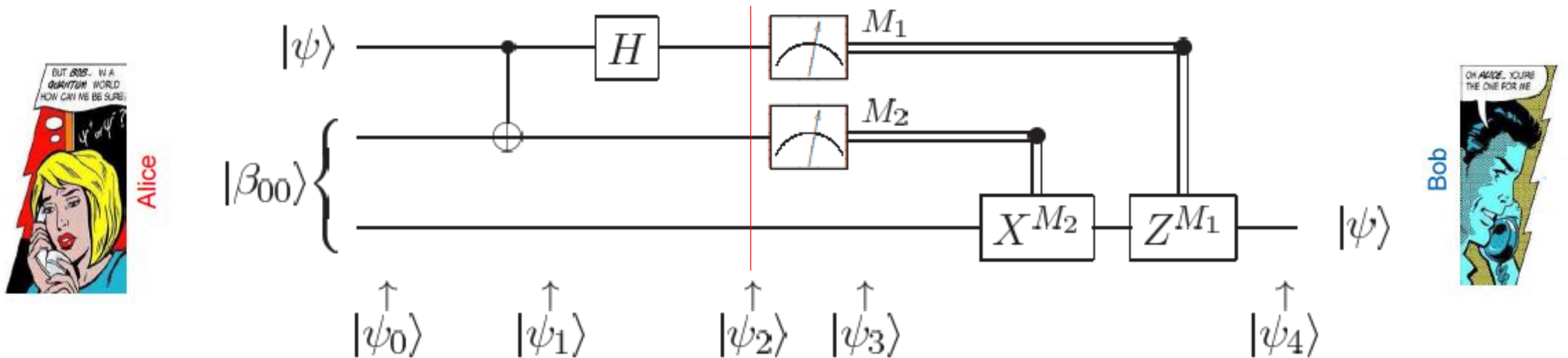


- Alice then sends the first qubit through a Hadamard gate, obtaining

$$|\psi_1\rangle = \frac{1}{\sqrt{2}} \left[\alpha |0_A\rangle (|0_A 0_B\rangle + |1_A 1_B\rangle) + \beta |1_A\rangle (|1_A 0_B\rangle + |0_A 1_B\rangle) \right]$$

$$|\psi_2\rangle = \frac{1}{2} \left[\alpha (|0_A\rangle + |1_A\rangle) (|0_A 0_B\rangle + |1_A 1_B\rangle) + \beta (|0_A\rangle - |1_A\rangle) (|1_A 0_B\rangle + |0_A 1_B\rangle) \right]$$

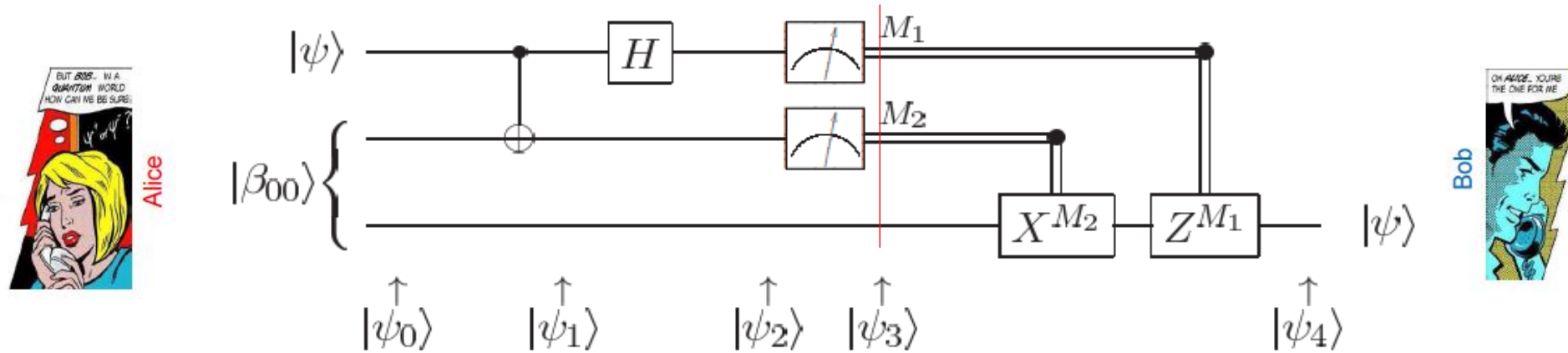
Quantum Teleportation



- $|\psi_2\rangle$ may be re-written in the following way, simply by regrouping terms:

$$\begin{aligned}
 |\psi_2\rangle = & \frac{1}{2} \left[|0_A 0_A\rangle (\alpha |0_B\rangle + \beta |1_B\rangle) + |0_A 1_A\rangle (\alpha |1_B\rangle + \beta |0_B\rangle) \right. \\
 & \left. + |1_A 0_A\rangle (\alpha |0_B\rangle - \beta |1_B\rangle) + |1_A 1_A\rangle (\alpha |1_B\rangle - \beta |0_B\rangle) \right]
 \end{aligned}$$

Quantum Teleportation



- Depending on Alice's measurement outcome, Bob's qubit will end up in one of the following four possible states

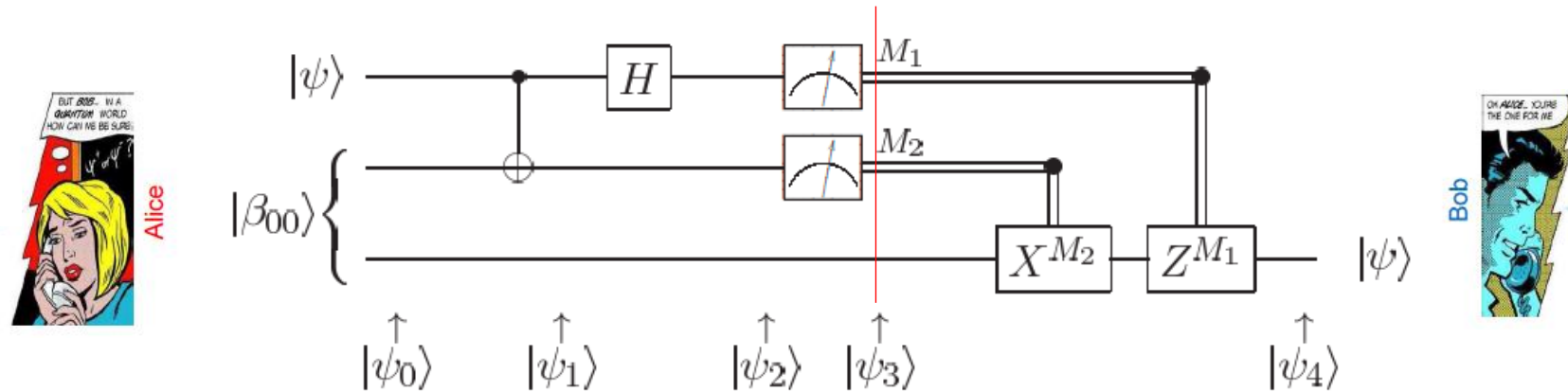
$$|\psi_2\rangle = \frac{1}{2} \left[|0_A 0_A\rangle (\alpha |0_B\rangle + \beta |1_B\rangle) + |0_A 1_A\rangle (\alpha |1_B\rangle + \beta |0_B\rangle) + |1_A 0_A\rangle (\alpha |0_B\rangle - \beta |1_B\rangle) + |1_A 1_A\rangle (\alpha |1_B\rangle - \beta |0_B\rangle) \right]$$

Measurement
by Alice

Post Measurement
state $|\psi_3\rangle$

Alice	Bob
$ 0_A 0_A\rangle$	$(\alpha 0_B\rangle + \beta 1_B\rangle)$
$ 0_A 1_A\rangle$	$(\alpha 1_B\rangle + \beta 0_B\rangle)$
$ 1_A 0_A\rangle$	$(\alpha 0_B\rangle - \beta 1_B\rangle)$
$ 1_A 1_A\rangle$	$(\alpha 1_B\rangle - \beta 0_B\rangle)$

Quantum Teleportation



To know which state it is in, Bob must be told the result of Alice's measurement

Alice

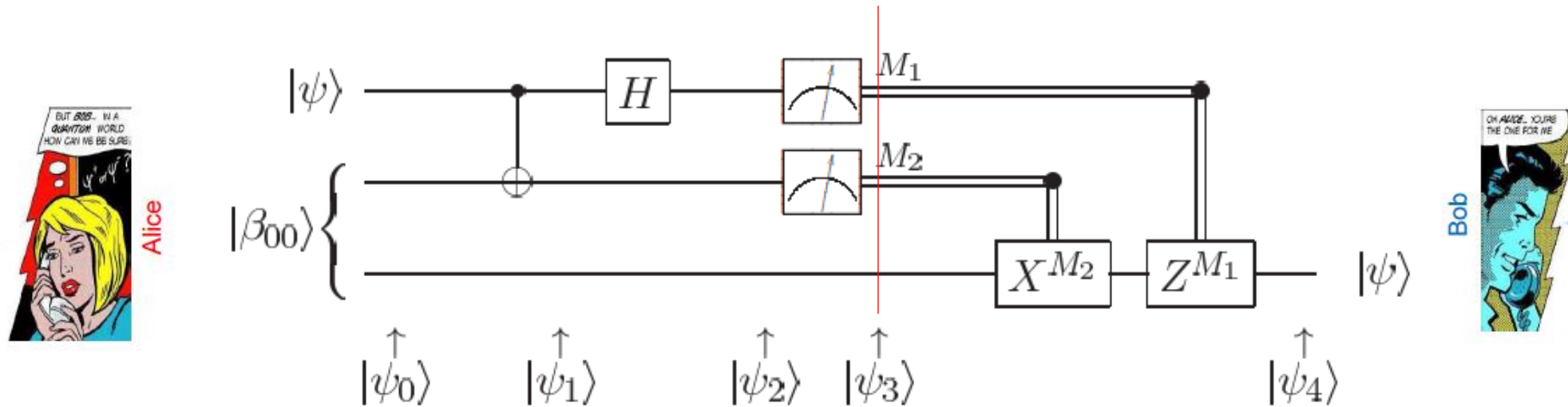
Bob

$$M_1 = 0_A, M_2 = 0_A \rightarrow \alpha|0_B\rangle + \beta|1_B\rangle$$

$$M_1 = 0_A, M_2 = 1_A \rightarrow \alpha|1_B\rangle + \beta|0_B\rangle$$

$$M_1 = 1_A, M_2 = 0_A \rightarrow \alpha|0_B\rangle - \beta|1_B\rangle$$

$$M_1 = 1_A, M_2 = 1_A \rightarrow \alpha|1_B\rangle - \beta|0_B\rangle$$



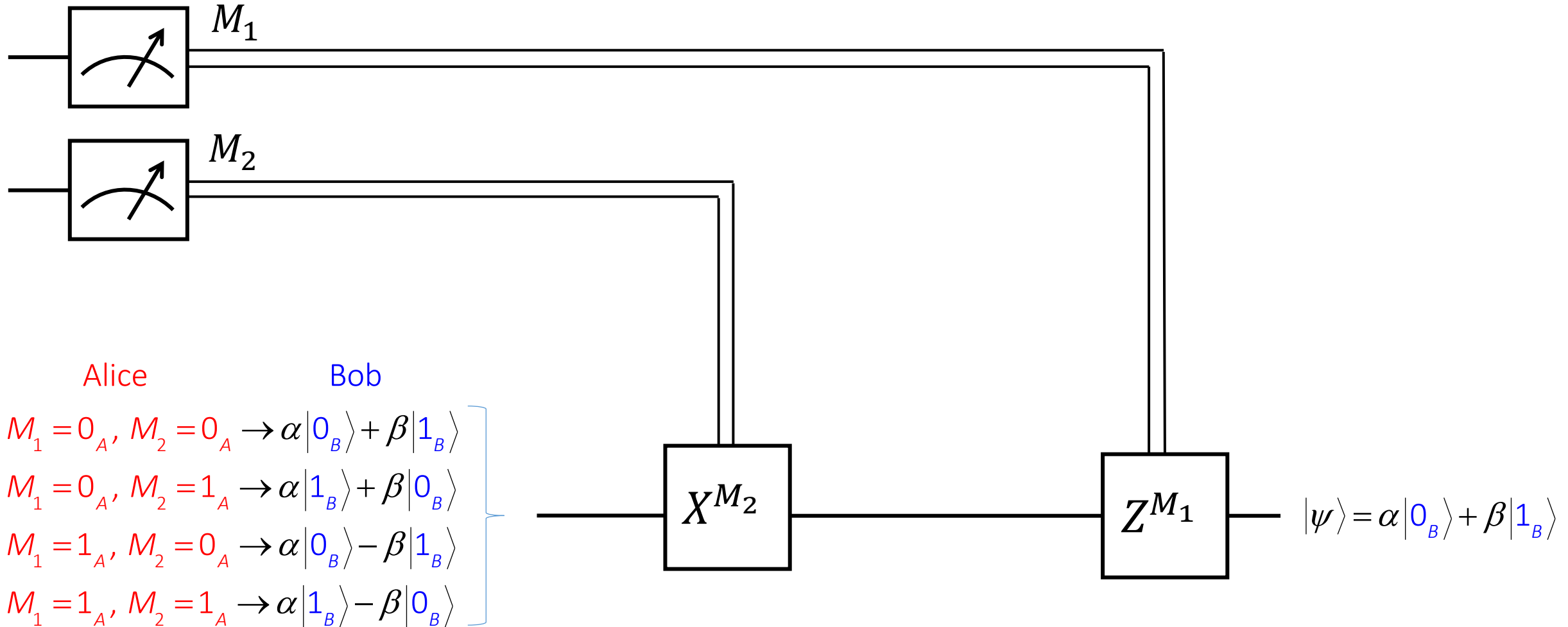
Once Bob has learned the measurement outcome, Bob can ‘fix up’ his state, recovering $|\psi\rangle$, by applying the appropriate quantum gate

Alice	Bob
$M_1 = 0_A, M_2 = 0_A \rightarrow \alpha 0_B\rangle + \beta 1_B\rangle$	
$M_1 = 0_A, M_2 = 1_A \rightarrow \alpha 1_B\rangle + \beta 0_B\rangle$	
$M_1 = 1_A, M_2 = 0_A \rightarrow \alpha 0_B\rangle - \beta 1_B\rangle$	
$M_1 = 1_A, M_2 = 1_A \rightarrow \alpha 1_B\rangle - \beta 0_B\rangle$	

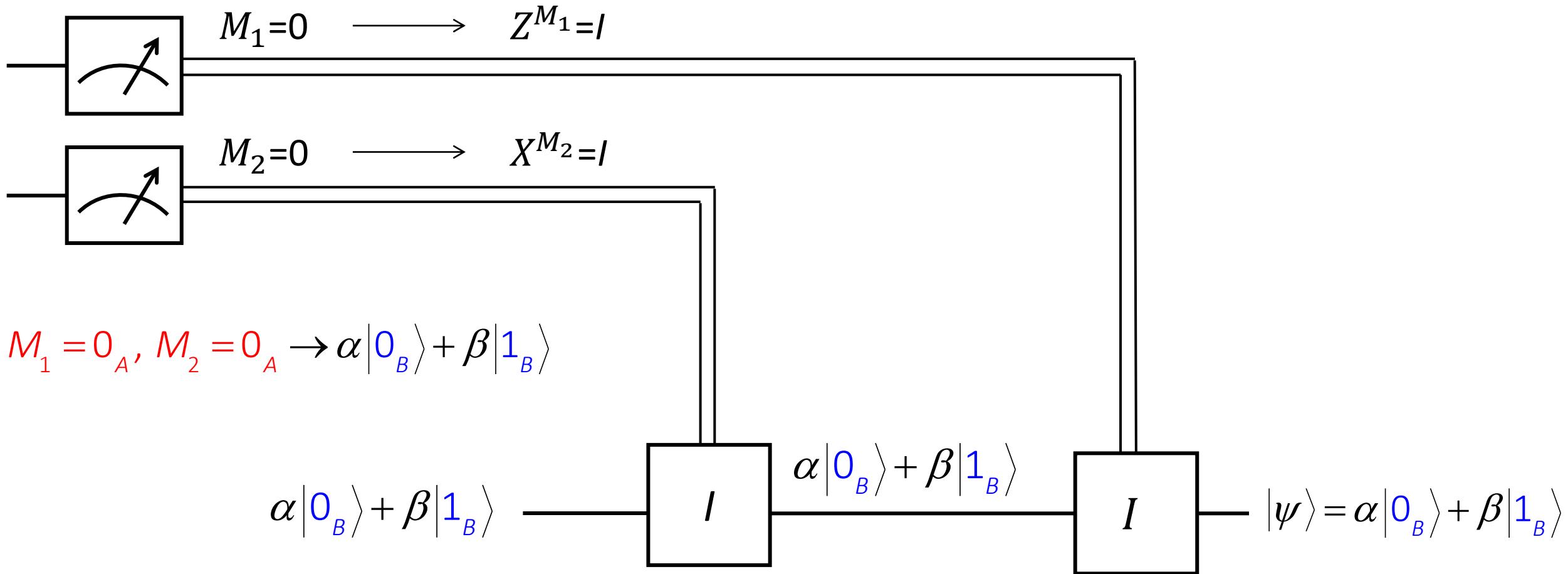
$$X^{M_2} = \begin{cases} X & \text{for } M_2 = 1 \\ I & \text{for } M_2 = 0 \end{cases}$$

$$Z^{M_1} = \begin{cases} Z & \text{for } M_1 = 1 \\ I & \text{for } M_1 = 0 \end{cases}$$

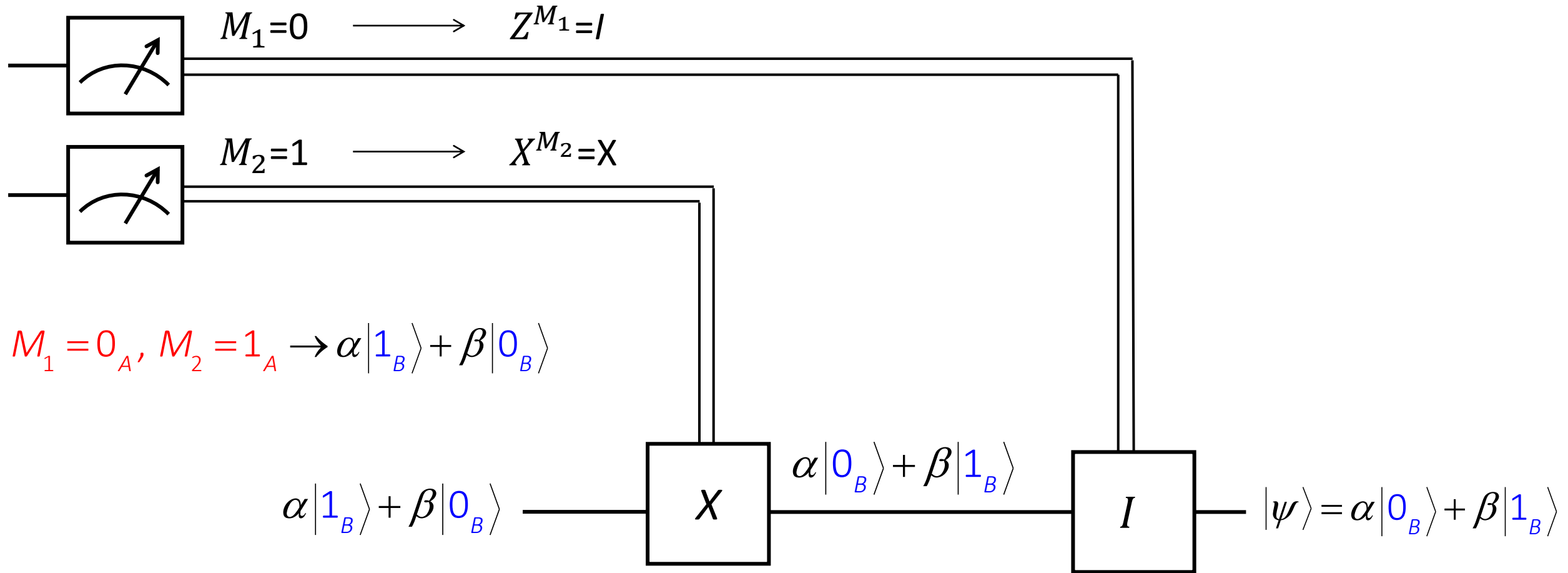
Quantum Teleportation



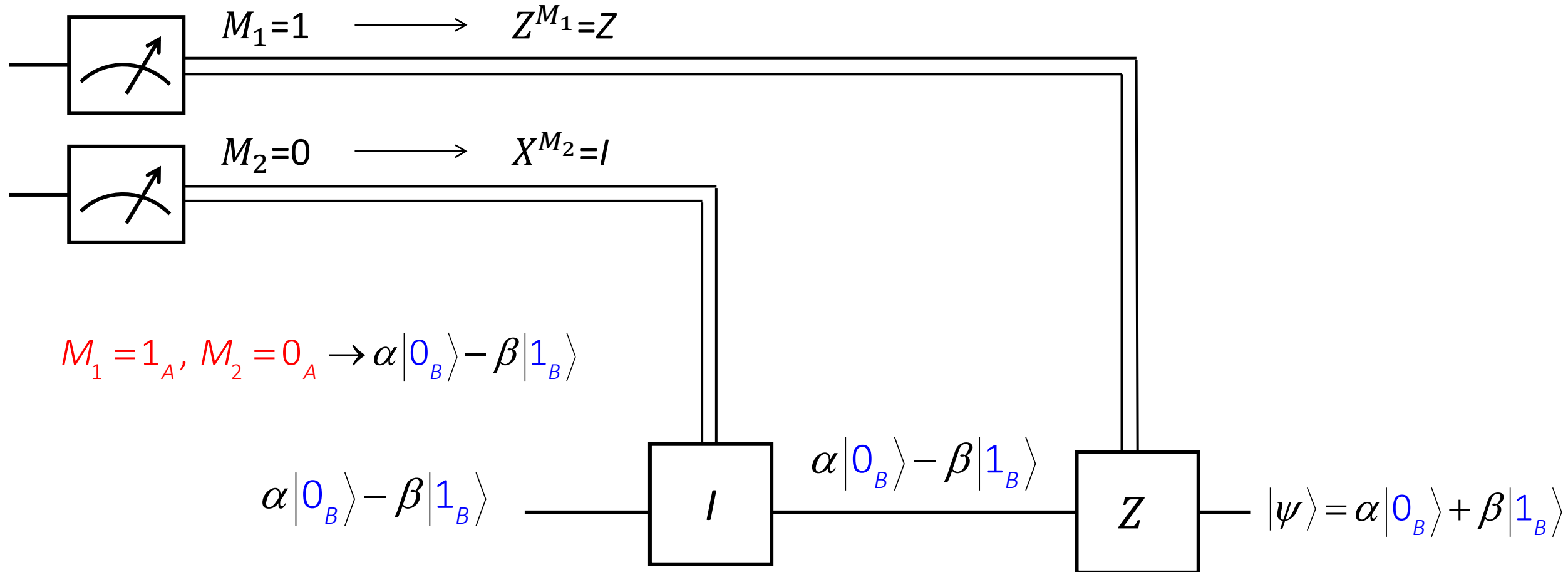
Quantum Teleportation



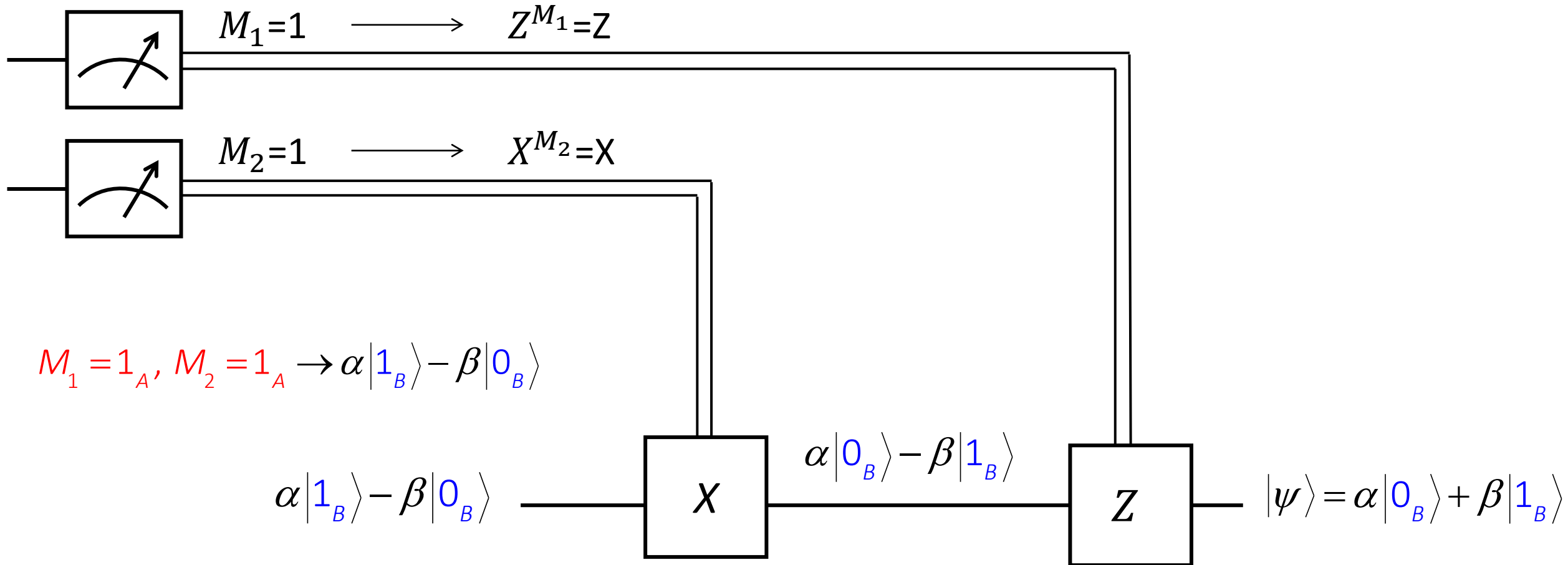
Quantum Teleportation



Quantum Teleportation



Quantum Teleportation



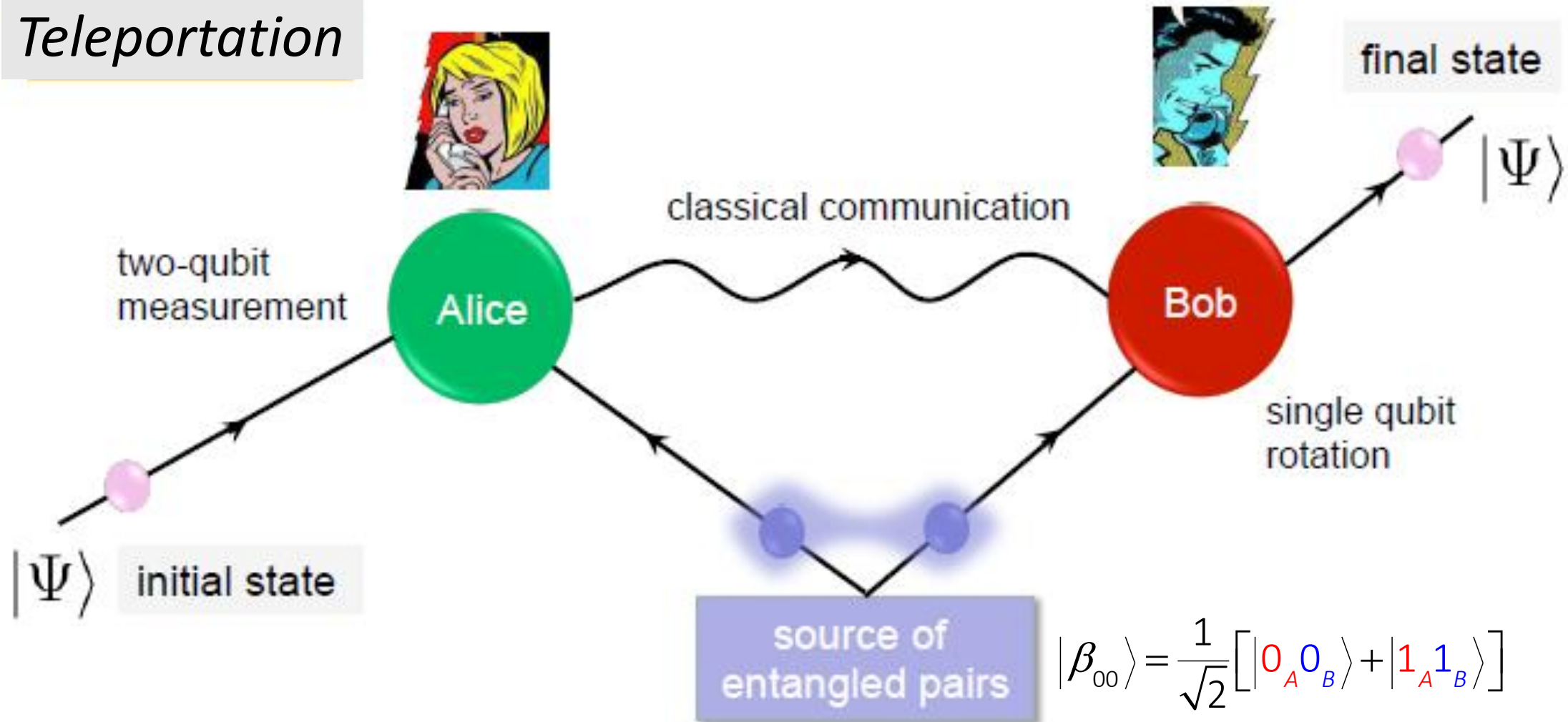
Quantum Teleportation Questions

- **First**, does teleportation allow one to transmit quantum states faster than light?
- No, because to complete the teleportation Alice must transmit her measurement result to Bob over a **classical communications channel**
- The classical channel is limited by the speed of light, resolving the apparent paradox

Quantum Teleportation Questions

- **Second**, teleportation appears to create a copy of the quantum state being teleported, in apparent violation of the no-cloning theorem
- This violation is only illusory since after the teleportation process only the target qubit is left in the state $|\psi\rangle$ and the original data qubit ends up in one of the computational basis states $|0\rangle$ or $|1\rangle$, depending upon the measurement result on the first qubit

Teleportation



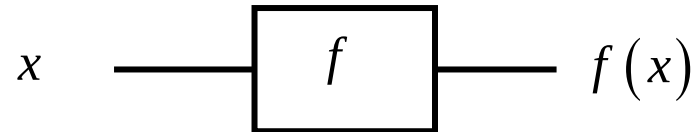
Deutsch Quantum Algorithm

Deutsch's Problem

- At this stage, we have acquired the concepts and tools needed to decide whether a given function has a certain property using the *Deutsch quantum algorithm*
- This computation cannot be solved as efficiently using any classical computer
- It's not a particularly useful calculation, mind you, because it's pretty contrived
- Nevertheless it highlights three key concepts of quantum computation: *quantum parallelism*, *quantum interference*, and the *phase kick-back*

The Problem: Is $f(x)$ Constant or Balanced?

- Suppose we are given a *black box* that computes some *function*, $f(x)$, even one that may be initially unknown to us



- The term *black box* suggests that we don't know what's on the inside or how it works
- Suppose we are dealing with classical function which takes a single classical bit in (one bit domain) and produces a single classical bit out (one bit range)

$$f : \{0,1\} \rightarrow \{0,1\}$$

The Problem: Is $f(x)$ Constant or Balanced?

- There are only a small number of functions that can act on the set $x \in \{0,1\}$ and give a single bit as output (i.e. one bit domain and range)
- For example, we could have the *identity* function

$$f(x) = \begin{cases} 0 & \text{if } x=0 \\ 1 & \text{if } x=1 \end{cases}$$

- Two more examples are the *constant* functions

$$f(0) = 0, f(1) = 0$$

$$f(0) = 1, f(1) = 1$$

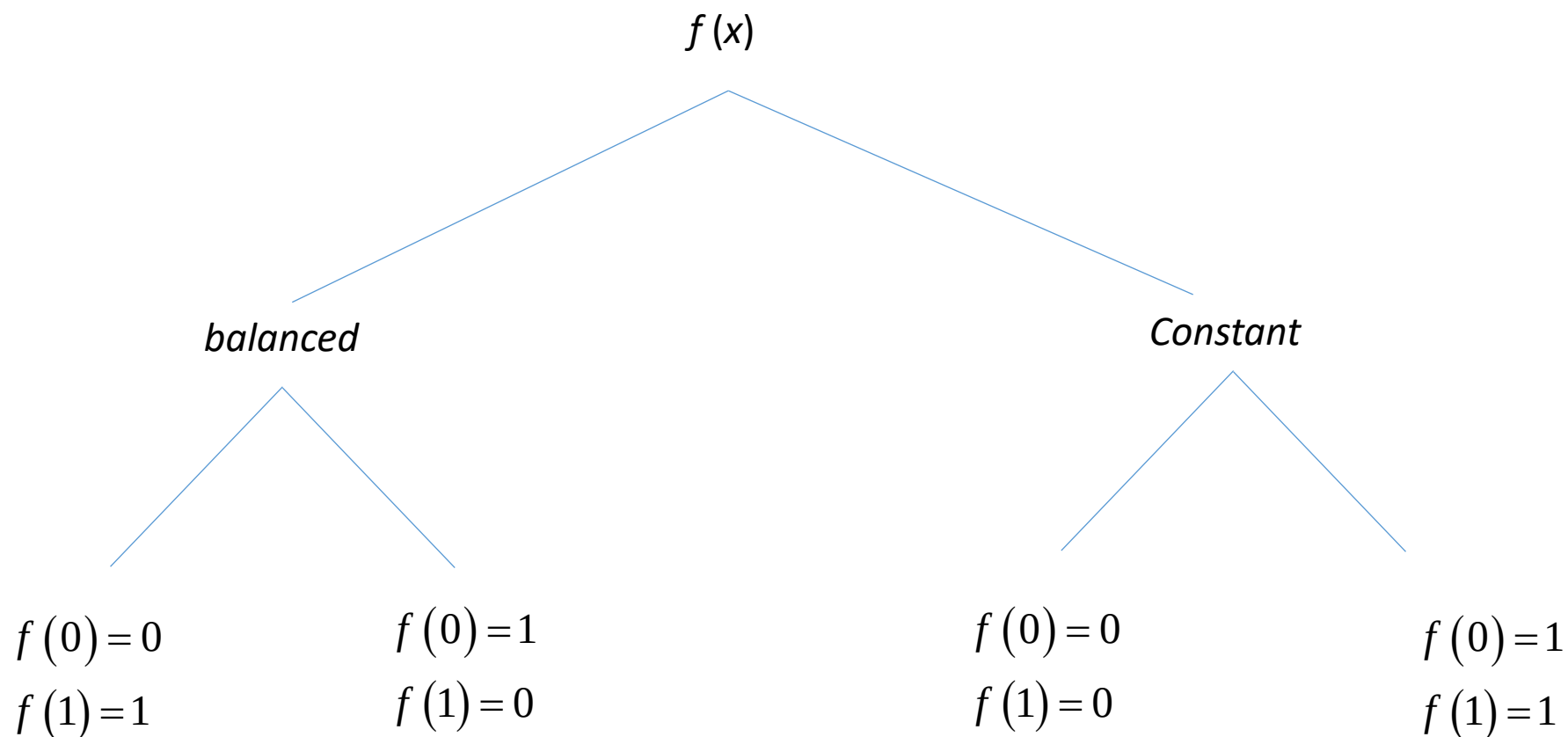
The Problem: Is $f(x)$ Constant or Balanced?

- The final example is the *bit flip* function

$$f(x) = \begin{cases} 1 & \text{if } x=0 \\ 0 & \text{if } x=1 \end{cases}$$

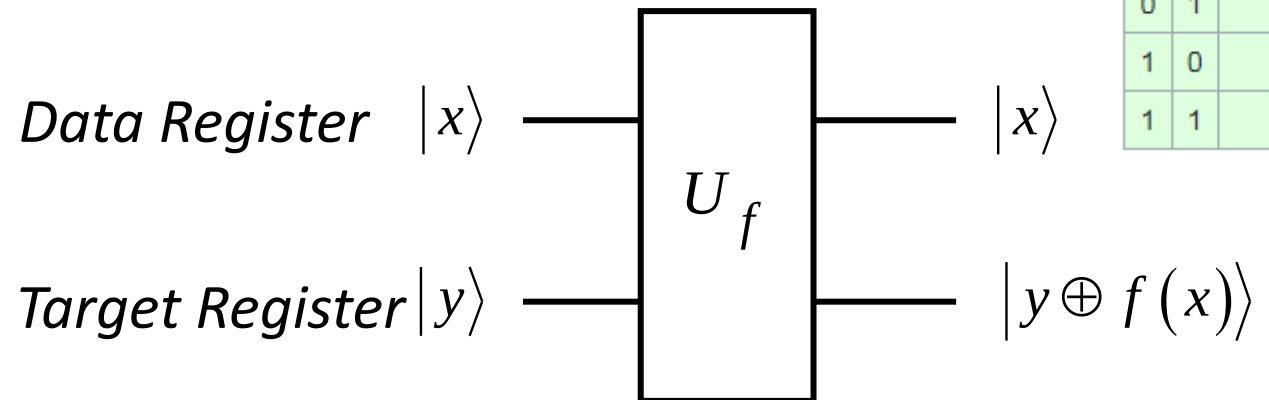
- The *identity* and *bit flip* functions are called *balanced* because the outputs are opposite for half the inputs
- So a **function on a single bit** can be *constant* or *balanced*

The Problem: Is $f(x)$ Constant or Balanced?



Embedding $f(x)$ in a Quantum Black-Box

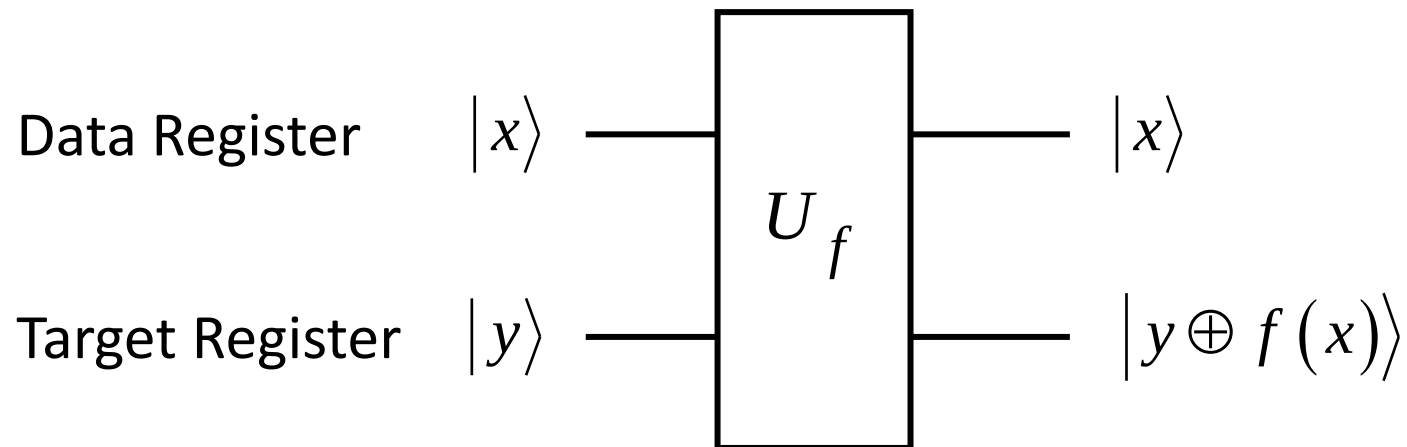
- A convenient way of *computing this function* on a quantum computer is to consider a two qubit quantum computer which starts in the state $|x\rangle|y\rangle$
- With an appropriate sequence of gates it is possible to transform $|x\rangle|y\rangle$ into $|x\rangle|y \oplus f(x)\rangle$, where $\oplus$ indicates addition modulo 2 (*XOR*)
- The first register is called the '*data*' register, and the second register the '*target*' register



Input		Output
A	B	A XOR B
0	0	0
0	1	1
1	0	1
1	1	0

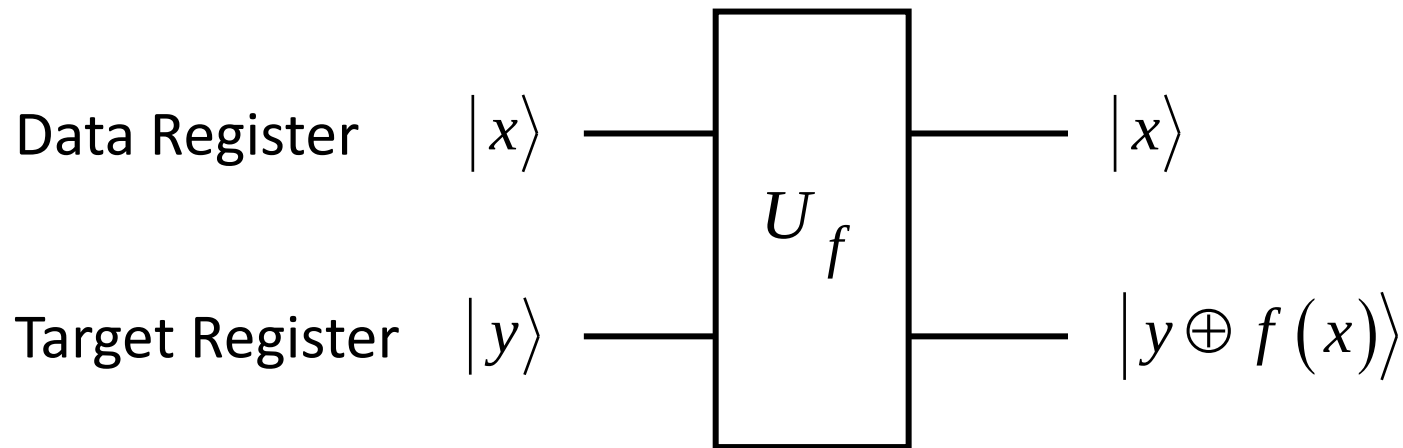
Quantum Oracle for $f(x)$

- We won't describe how this works but, instead, take it as a given and call the new, larger circuit " U_f " the **quantum oracle for f**
- Its action on CBS basis and its circuit diagram are defined by



Quantum Oracle for $f(x)$

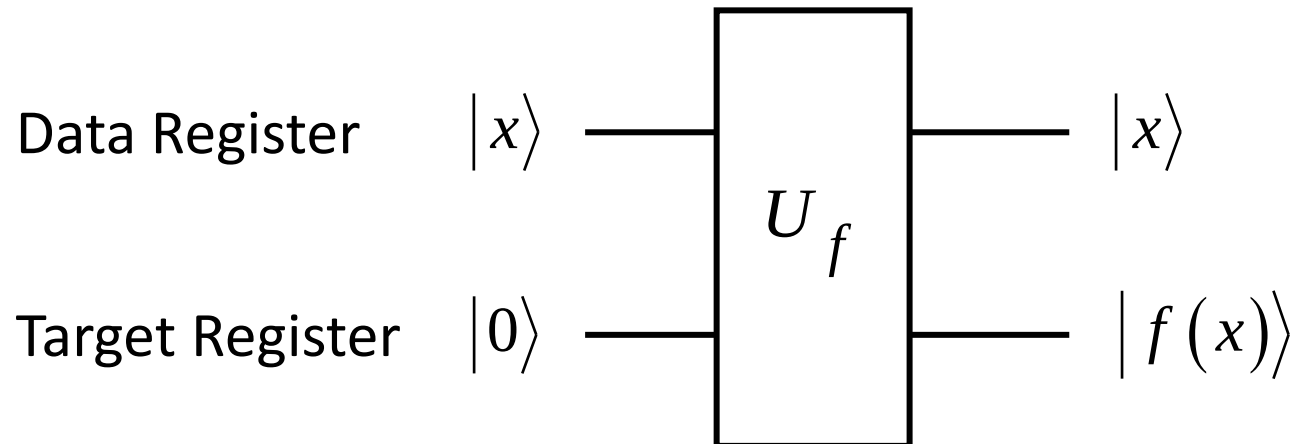
- First, notice that the output of the target register is a CBS; inside the ket we are *XOR*-ing ($\oplus$) two classical binary values y and $f(x)$, producing another binary value which, in turn, defines a CBS ket: either $|0\rangle$ or $|1\rangle$



Notice that output of U_f for a CBS is always a separable state $|y \oplus f(x)\rangle$; since $y \oplus f(x)$ is always either 0 or 1

Quantum Oracle for $f(x)$

- U_f computes $f(x)$
- This is a consequence of the circuit definition, because if we plug, $0 \rightarrow y$ we get



Deutsch's Problem

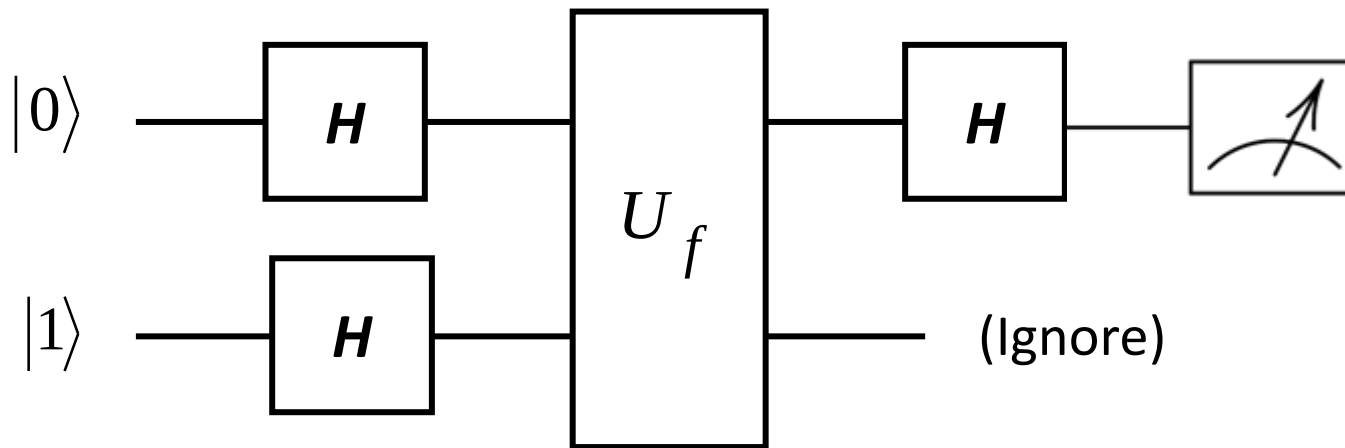
- Given an *unknown function* with one-bit domain and range, that we are told is either *balanced* or *constant*, determine in *one query* of the quantum oracle, whether f is *constant*
- Notice that we are not asking to determine the exact function, just which *category* it belongs to
- **Even so, we cannot do it classically in a single query**

Deutsch's Algorithm

- The algorithm consists of building a circuit and measuring the *Data Register* - *once*
- Our conclusion about f is determined by the result of the measurement
- If we get a 0 the function is *constant*, if we get 1 the function is *balanced*
- We will have gotten an answer about f with only one query of the oracle and thereby obtained a x2 improvement over a classical algorithm

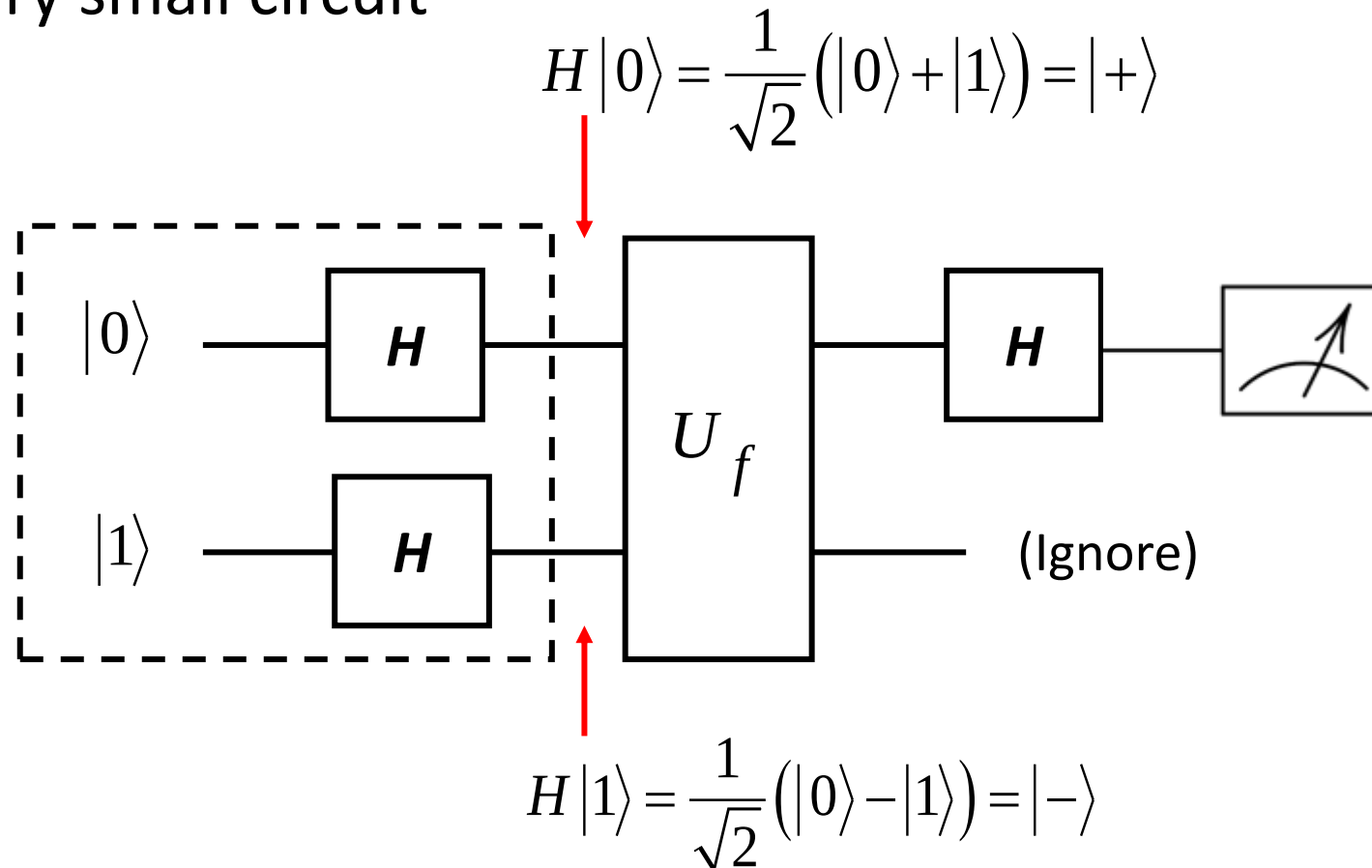
The Circuit

- We combine the quantum oracle for f with a few *Hadamard* gates as shown in the following small circuit



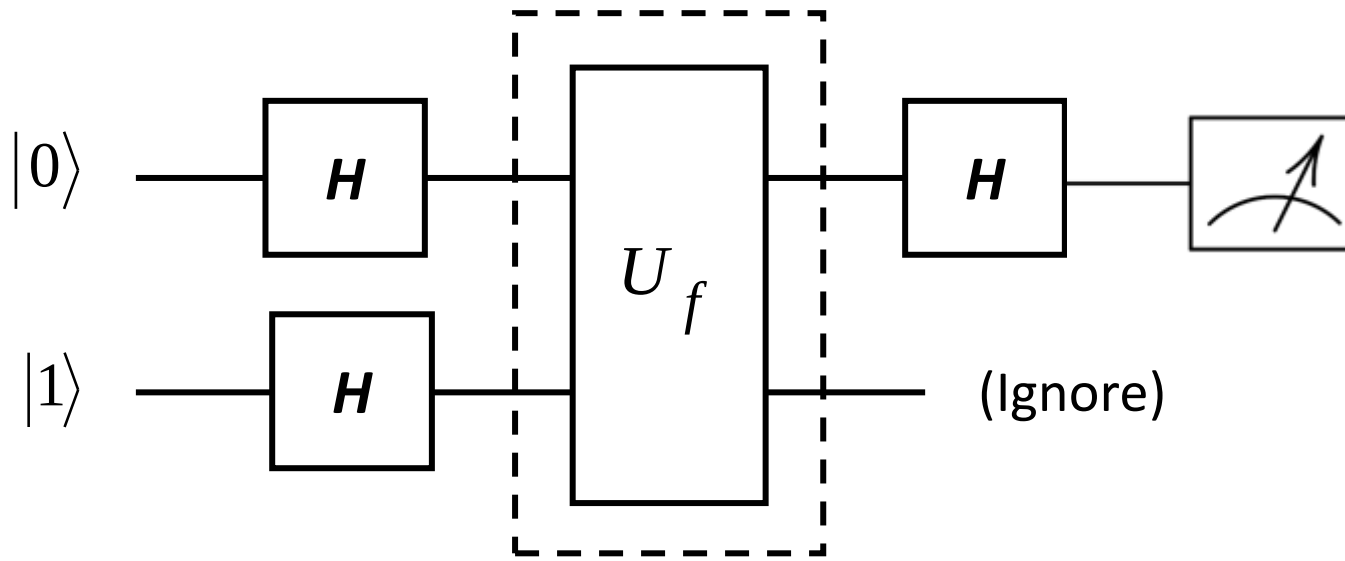
Preparing the Oracle's Input

- We combine the quantum oracle for f with a few *Hadamard* gates in a very small circuit



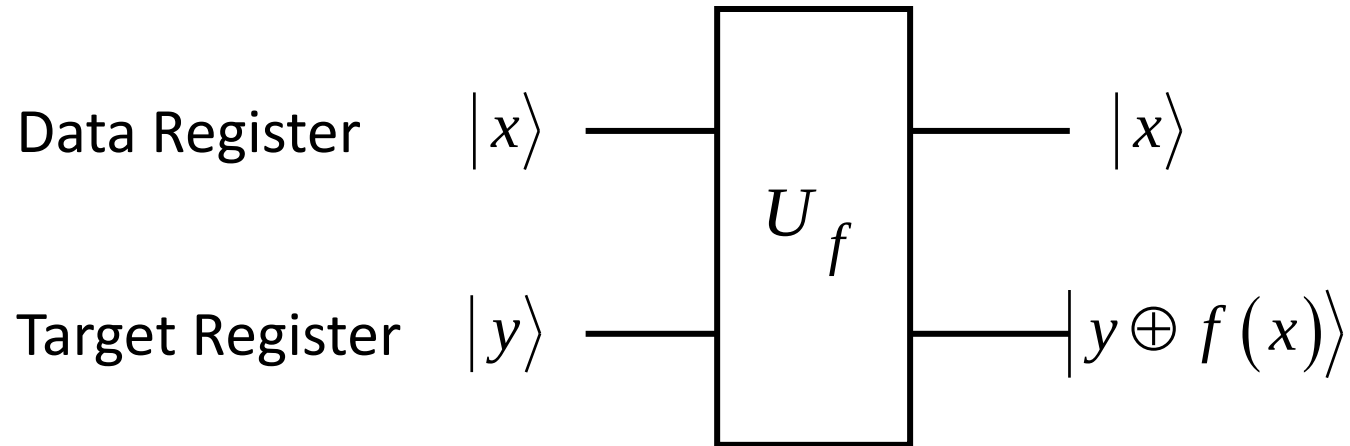
Analyzing the Oracle

- The real understanding of how the algorithm works comes by analyzing the kernel of the circuit, the oracle (in the dashed-box),



Analyzing the Oracle

- Step 1. CBS Into Both Channels



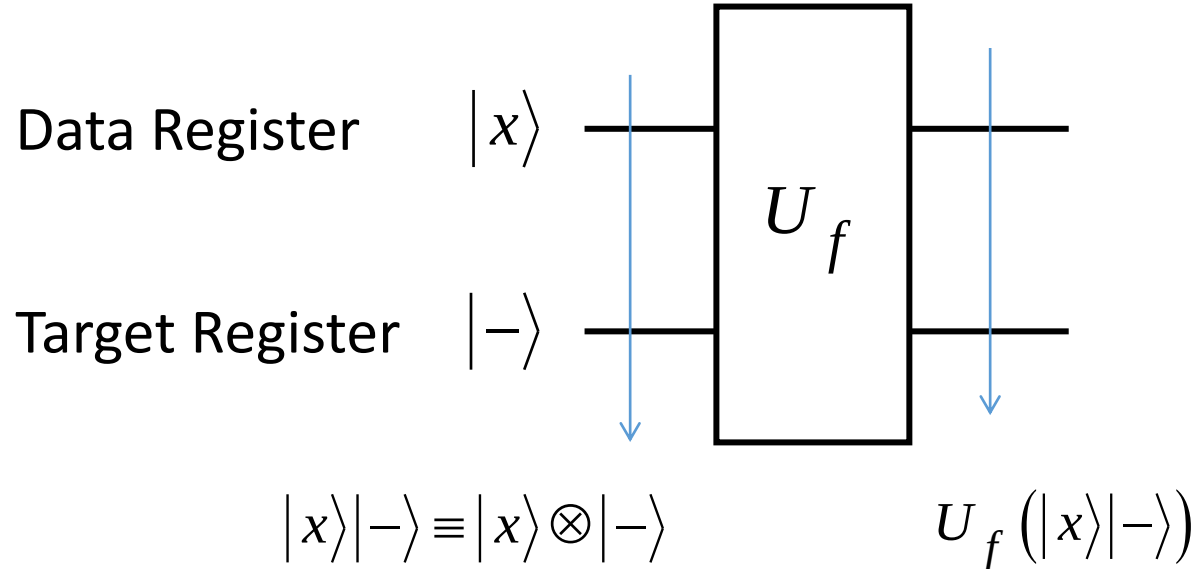
- Algebraically

$$|x\rangle|y\rangle \rightarrow U_f(|x\rangle|y\rangle) = |x\rangle|y \oplus f(x)\rangle \quad \Rightarrow$$

Both, input and output of the quantum oracle are separable

Analyzing the Oracle

- **Step 2.** CBS Into Data and Superposition into Target
- We stick with a CBS $|x\rangle$ going into the data register, but now allow the superposition $|-\rangle$ go into the target register



Analyzing the Oracle

- Extend the above linearly,

$$U_f(|x\rangle|-\rangle) = U_f\left(|x\rangle\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)\right) = \frac{1}{\sqrt{2}}U_f(|x\rangle(|0\rangle - |1\rangle)) = \frac{1}{\sqrt{2}}U_f(|x\rangle|0\rangle - |x\rangle|1\rangle)$$

$$= \frac{1}{\sqrt{2}}(U_f(|x\rangle|0\rangle) - U_f(|x\rangle|1\rangle)) = \frac{1}{\sqrt{2}}(|x\rangle|0 \oplus f(x)\rangle - |x\rangle|1 \oplus f(x)\rangle)$$

$$= \frac{1}{\sqrt{2}}(|x\rangle|f(x)\rangle - |x\rangle|\overline{f(x)}\rangle)$$

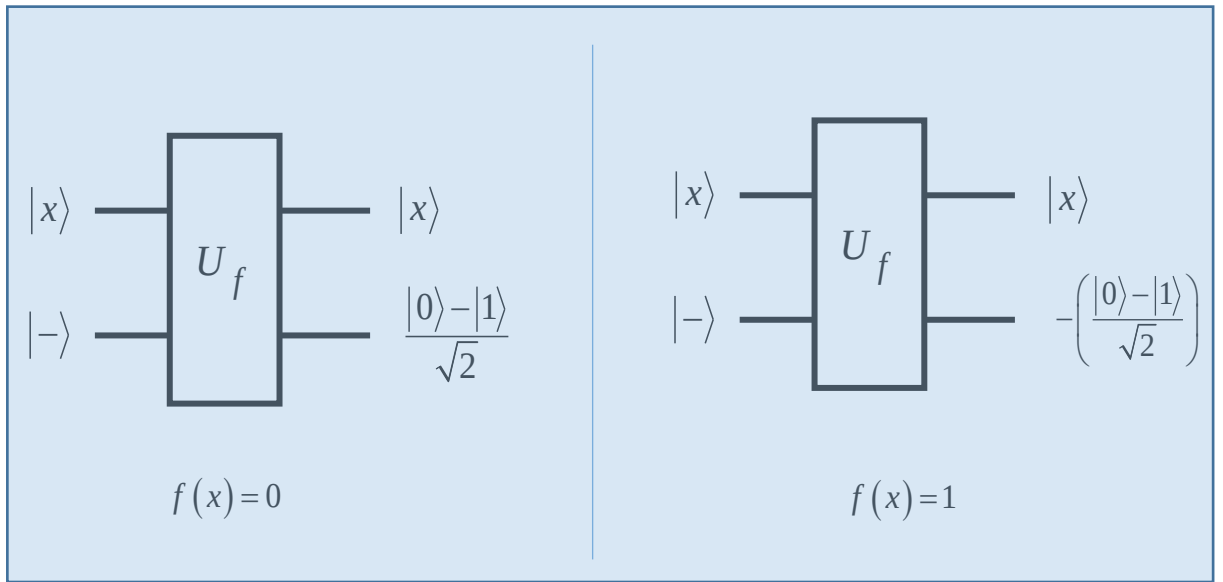
$$= |x\rangle\left(\frac{|f(x)\rangle - |\overline{f(x)}\rangle}{\sqrt{2}}\right)$$

Analyzing the Oracle

- This amounts to

$$U_f(|x\rangle|-\rangle) = |x\rangle \begin{cases} \frac{|0\rangle - |1\rangle}{\sqrt{2}} & \text{when } f(x) = 0 \\ -\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) & \text{when } f(x) = 1 \end{cases}$$

$$= |x\rangle (-1)^{f(x)} \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

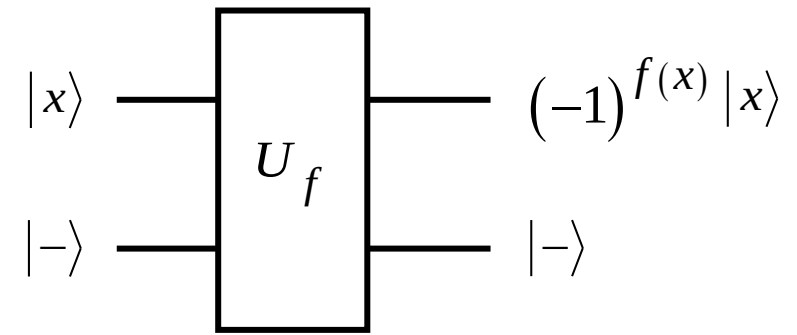


Analyzing the Oracle

Since it's a scalar, $(-1)^{f(x)}$ can be moved to the left and be attached to the Data Register's $|x\rangle$, a mere rearrangement of the terms

$$U_f(|x\rangle|-\rangle) = (-1)^{f(x)}|x\rangle\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) = \left((-1)^{f(x)}|x\rangle\right)|-\rangle$$

and we have successfully *(like magic)* moved all of the information about $f(x)$ from the Target Register to the Data Register, where it appears as an *overall phase factor* in the scalar's exponent



Analyzing the Oracle

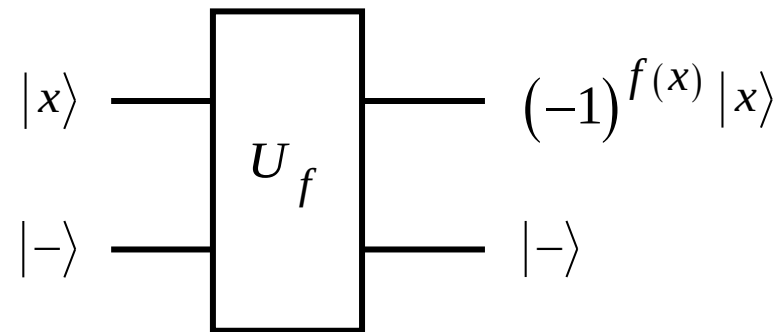
Let's pause to summarize what we have accomplished so far

- We have proven that $|x\rangle|-\rangle$ is an eigenvector of U_f with eigenvalue $(-1)^{f(x)}$ for $x = 0, 1$

$$U_f(|x\rangle|-\rangle) = \left((-1)^{f(x)}|x\rangle\right)|-\rangle = (-1)^{f(x)}(|x\rangle|-\rangle)$$

Analyzing the Oracle

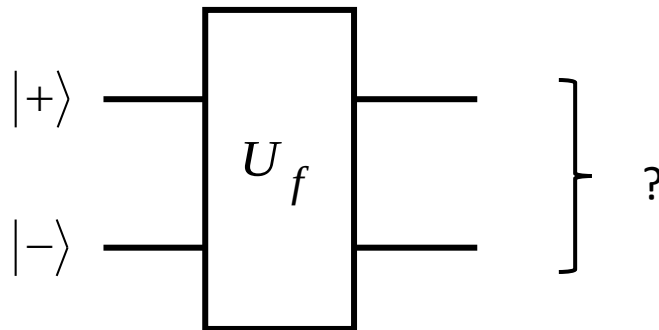
- The information about $f(x)$ is encoded *“kicked-back”* in the Data Register's output
- That's where we plan to look for it in the coming step
- Viewed this way, the Target Register retains *no useful* information



Analyzing the Oracle

Step 3. Superpositions into Both Registers

- Finally, we want the state $|+\rangle$ to go into the data register so we can process both $f(0)$ and $f(1)$ in a single pass
- The effect is to present the separable $|+\rangle \otimes |-\rangle$ to the oracle and see what comes out



Analyzing the Oracle

Applying linearity to the last result we get

$$U_f(|+\rangle|-\rangle) = U_f\left(\frac{|0\rangle+|1\rangle}{\sqrt{2}}|-\rangle\right) = U_f\left(\frac{|0\rangle|-\rangle+|1\rangle|-\rangle}{\sqrt{2}}\right) = \frac{U_f(|0\rangle|-\rangle) + U_f(|1\rangle|-\rangle)}{\sqrt{2}}$$

$$= \frac{(-1)^{f(0)}|0\rangle|-\rangle + (-1)^{f(1)}|1\rangle|-\rangle}{\sqrt{2}}$$

$$= \frac{(-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle}{\sqrt{2}}|-\rangle$$

$$U_f(|x\rangle|-\rangle) = \left((-1)^{f(x)}|x\rangle\right)|-\rangle$$



Analyzing the Oracle

$$\frac{(-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle}{\sqrt{2}}|-\rangle \quad \leftarrow$$

This is a remarkable state! The different terms contain information about both $f(0)$ and $f(1)$; it is almost as if we have evaluated $f(x)$ for two values of x *simultaneously*, a feature known as **QUANTUM PARALLELISM**

Thus, by combining the *phase kick-back* with *quantum parallelism*, we've managed to get an expression containing both $f(0)$ and $f(1)$ in the Data Register

Analyzing the Oracle

- Unlike *classical parallelism*, where *multiple* circuits each built to compute $f(x)$ are executed simultaneously, here a *single* $f(x)$ circuit is employed to evaluate the function for *multiple values of x simultaneously*, by exploiting the ability of a qubit to be in **superpositions of different states**

Analyzing the Oracle

- **Question:** what is the difference between the *balanced case* ($f(0) \neq f(1)$) and the *constant case* ($f(0) = f(1)$)?
- **Answer:** when constant, the two terms in the numerator have the same sign and when balanced, they have different signs, to wit

$$\frac{(-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle}{\sqrt{2}}|-\rangle \longrightarrow U_f(|+\rangle|-\rangle) = \begin{cases} (\pm 1) \frac{|0\rangle + |1\rangle}{\sqrt{2}}|-\rangle & \text{if } f(0) = f(1) \\ (\pm 1) \frac{|0\rangle - |1\rangle}{\sqrt{2}}|-\rangle & \text{if } f(0) \neq f(1) \end{cases}$$

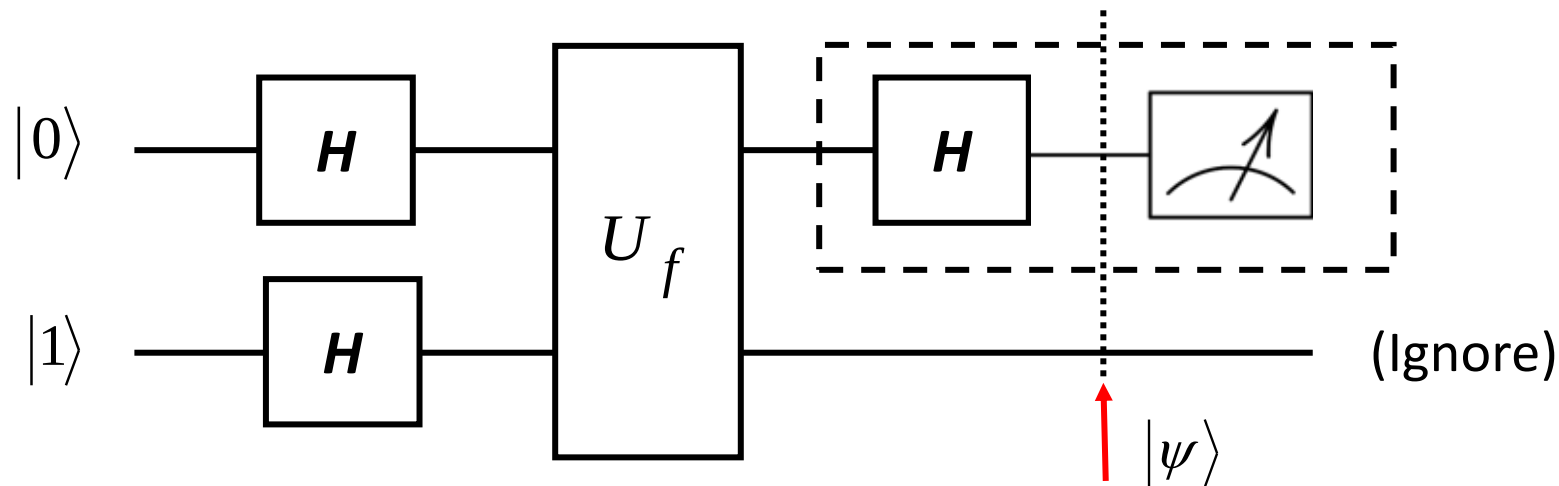
Analyzing the Oracle

- We don't care about a possible overall phase factor or (-1) in front of all this since it's a unit scalar in a state space
- Thus, in the following we dump it

$$\frac{(-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle}{\sqrt{2}}|-\rangle \quad \longrightarrow \quad U_f(|+\rangle|-\rangle) = \begin{cases} \frac{|0\rangle + |1\rangle}{\sqrt{2}}|-\rangle & \text{if } f(0) = f(1) \\ \frac{|0\rangle - |1\rangle}{\sqrt{2}}|-\rangle & \text{if } f(0) \neq f(1) \end{cases}$$

Measurement

- The final Hadamard gate on the Target Register thus gives



$$\frac{|0\rangle + |1\rangle}{\sqrt{2}} |-\rangle \rightarrow |\psi\rangle = (H \otimes I) \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} |-\rangle \right) = \left(H \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \right) \otimes (I |-\rangle) = |0\rangle |-\rangle \quad \text{if } f(0) = f(1)$$

$$\frac{|0\rangle - |1\rangle}{\sqrt{2}} |-\rangle \rightarrow |\psi\rangle = (H \otimes I) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} |-\rangle \right) = \left(H \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \right) \otimes (I |-\rangle) = |1\rangle |-\rangle \quad \text{if } f(0) \neq f(1)$$

Measurement

- Realizing that

$$f(0) \oplus f(1) = \begin{cases} 0 & \text{if } f(0) = f(1) \\ 1 & \text{if } f(0) \neq f(1) \end{cases}$$

we can rewrite the above result

$$|\psi\rangle = |f(0) \oplus f(1)\rangle \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right] = |f(0) \oplus f(1)\rangle |-\rangle$$

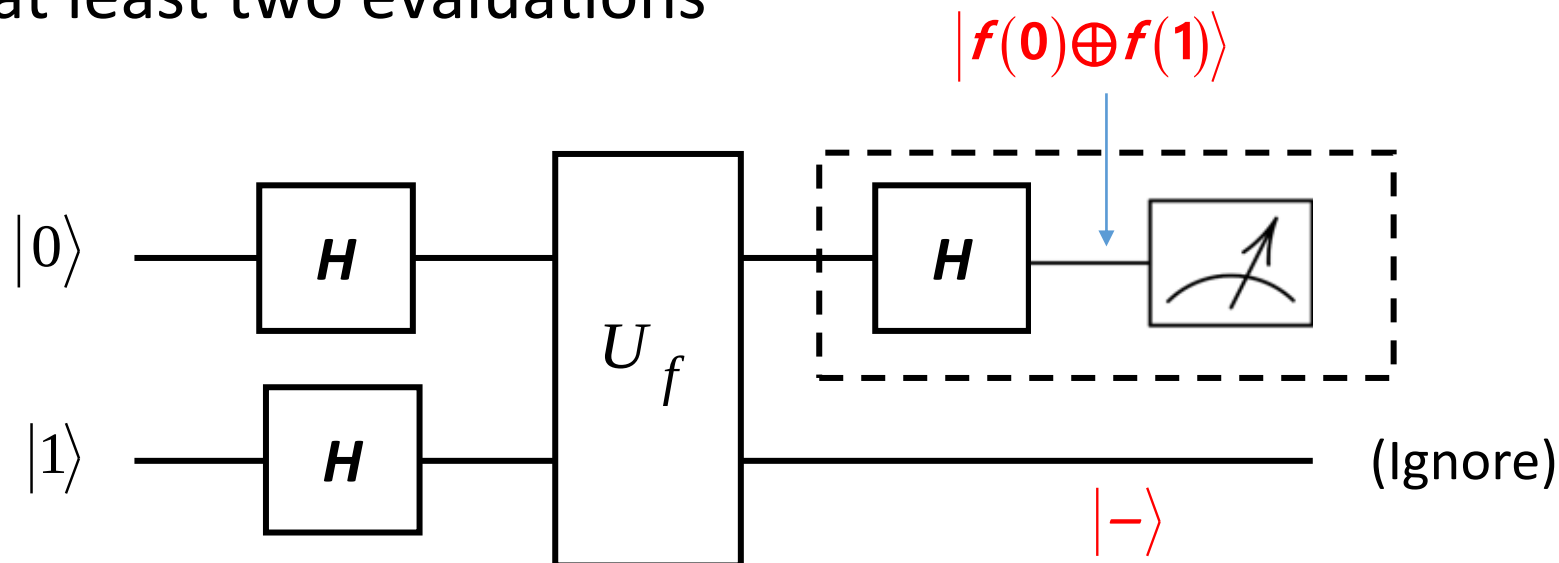
so by measuring the *first qubit* we may determine

$$f(0) \oplus f(1)$$

We only care about the first qubit, since the second qubit will always collapse to $|-\rangle$

Measurement

- This is very interesting indeed: the quantum circuit has given us the ability to determine a *global property* of $f(x)$, namely $f(0) \oplus f(1)$, using only *one* evaluation of $f(x)$!
- This is faster than is possible with a classical apparatus, which would require at least two evaluations



Measurement

- In a quantum computer it is possible for the two alternatives to **interfere** with one another to yield some global property of the function f , by using something like the Hadamard gate to recombine the different alternatives, as was done in Deutsch's algorithm
- The essence of the design of many quantum algorithms is that a clever choice of function and final transformation allows efficient determination of useful global information about the function - information which cannot be attained quickly on a classical computer

The Deutsch–Jozsa Algorithm

The Deutsch–Jozsa Algorithm

- The *Deutsch–Jozsa algorithm* solves a problem that is a straightforward generalization of the problem solved by the Deutsch algorithm
- As with the *Deutsch algorithm*, we are given a reversible circuit implementing an unknown function f , but this time f is a function from *n -bit strings* $x = x_1x_2x_3 \cdots x_n$ to *a single bit*. That is,

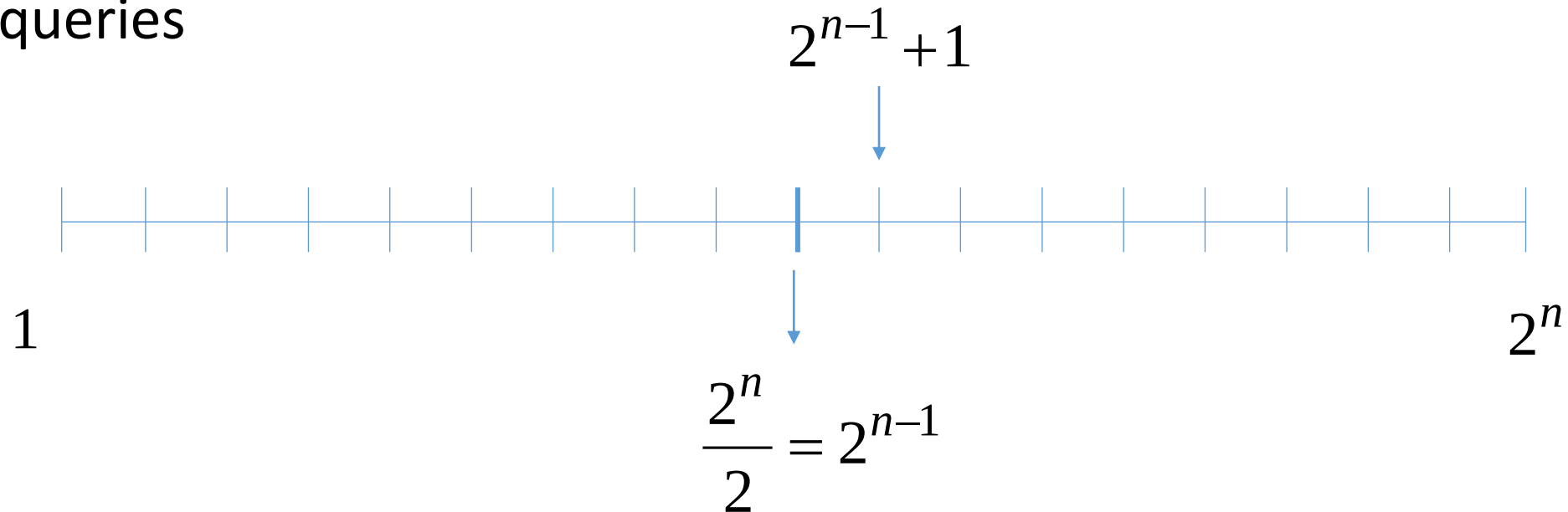
$$f : \{x_1, x_2, x_3, \dots, x_n\} \rightarrow \{0, 1\} \quad \text{where} \quad x_i \in \{0, 1\}, \quad i \in \{1, 2, \dots, n\}$$

The Deutsch–Jozsa Algorithm

- We are also given the promise that
 - f is either *constant* (meaning $f(x)$ is the same for all x), or
 - f is *balanced* (meaning $f(x) = 0$ for exactly half of the input strings x , and $f(x) = 1$ for the other half of the inputs)
- The problem here is to determine whether f is *constant*, or *balanced*, by making queries to the circuit for f

The Deutsch–Jozsa Algorithm

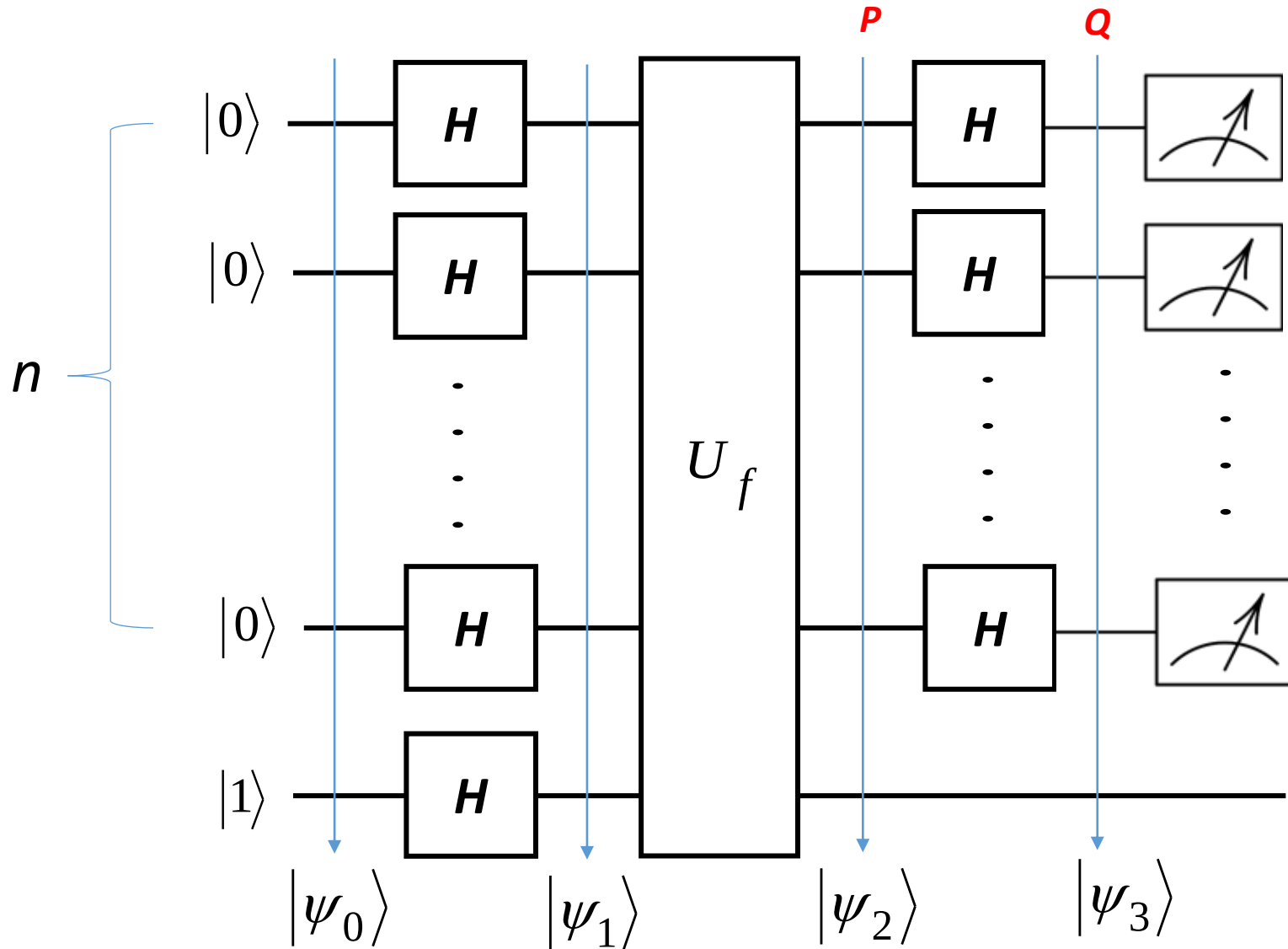
- Consider solving this problem by a classical algorithm
- In the worst case, using a classical algorithm we cannot decide with certainty whether f is constant or balanced using any less than $2^{n-1} + 1$ queries



The Deutsch–Jozsa Algorithm

- The Deutsch-Jozsa algorithm is very similar to Deutsch's algorithm, but we now have *n qubits* plus the *target qubit*
- These *n qubits* are initially each in the $|0\rangle$ state, and we apply Hadamards to put them in a superposition of all n -bit strings
- Then we query the oracle on this superposition
- Finally, we apply Hadamards to all the qubits to create a state that we measure, and whose measurement outcome allows us to distinguish whether the function is constant or balanced

The Deutsch–Jozsa Algorithm



To determine if f is constant or balanced, we measure the n **qubits**, and if we get $|000\cdots 0\rangle$, the function is **constant**, and if we get anything else, the function is **balanced**

The Deutsch–Jozsa Algorithm

- As we did for the Deutsch algorithm, we follow the state through the circuit. Initially the state is

$$|\psi_0\rangle = |0\rangle^{\otimes n} |1\rangle$$

- After the Hadamard transform on the query register and the Hadamard gate on the answer register we have

$$|\psi_1\rangle = H^{\otimes n+1} \left(|0\rangle^{\otimes n} |1\rangle \right) = \left(H^{\otimes n} \otimes H \right) |0\rangle^{\otimes n} \otimes |1\rangle = \left(H^{\otimes n} |0\rangle^{\otimes n} \right) \otimes \left(H |1\rangle \right)$$

The Deutsch–Jozsa Algorithm

- Consider the action of an n -qubit Hadamard transformation on the state $|0\rangle^{\otimes n}$

$$H^{\otimes n} |0\rangle^{\otimes n} = \underbrace{(H \otimes H \otimes \dots \otimes H)}_n \underbrace{(|0\rangle \otimes |0\rangle \otimes \dots \otimes |0\rangle)}_n = \underbrace{(H|0\rangle) \otimes (H|0\rangle) \otimes \dots \otimes (H|0\rangle)}_n$$

$$= \left(\frac{1}{\sqrt{2}}\right)^n \underbrace{(|0\rangle + |1\rangle) \otimes (|0\rangle + |1\rangle) \otimes \dots \otimes (|0\rangle + |1\rangle)}_n$$

$$\{0,1\}^n = \underbrace{\{0,1\} \times \{0,1\} \times \dots \times \{0,1\}}_n$$

$$= \left(\frac{1}{\sqrt{2}}\right)^n \underbrace{|00\dots 00\rangle \otimes |00\dots 01\rangle \otimes |00\dots 10\rangle \otimes \dots \otimes |11\dots 10\rangle |11\dots 11\rangle}_n = \left(\frac{1}{\sqrt{2}}\right)^n \sum_{x \in \{0,1\}^n} |x\rangle$$

The Deutsch–Jozsa Algorithm

- This is a very common and useful way of writing this state; the n -qubit Hadamard gate acting on the n -qubit state of all zeros gives a superposition of all n -qubit basis states, all with the same amplitude (called an ‘equally weighted superposition’) $1/\sqrt{2^n}$
- So the state immediately after the first $H^{\otimes n}$ in the Deutsch–Jozsa algorithm is

$$|\psi_1\rangle = \left(\frac{1}{\sqrt{2}}\right)^n \sum_{x \in \{0,1\}^n} |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

The Deutsch–Jozsa Algorithm

- Now consider the state immediately after the U_f gate. The state is

$$\begin{aligned} |\psi_2\rangle &= \left(\frac{1}{\sqrt{2}}\right)^n U_f \left(\sum_{x \in \{0,1\}^n} |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \right) = \left(\frac{1}{\sqrt{2}}\right)^n \sum_{x \in \{0,1\}^n} U_f |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \\ &= \left(\frac{1}{\sqrt{2}}\right)^n \sum_{x \in \{0,1\}^n} U_f \left(\frac{|x\rangle|0\rangle - |x\rangle|1\rangle}{\sqrt{2}} \right) = \left(\frac{1}{\sqrt{2}}\right)^n \sum_{x \in \{0,1\}^n} \left(\frac{U_f(|x\rangle|0\rangle) - U_f(|x\rangle|1\rangle)}{\sqrt{2}} \right) \\ &= \left(\frac{1}{\sqrt{2}}\right)^n \sum_{x \in \{0,1\}^n} \left(\frac{|x\rangle|0 \oplus f(x)\rangle - |x\rangle|1 \oplus f(x)\rangle}{\sqrt{2}} \right) = \left(\frac{1}{\sqrt{2}}\right)^n \sum_{x \in \{0,1\}^n} (-1)^{f(x)} |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \end{aligned}$$

The Deutsch–Jozsa Algorithm

- To facilitate our analysis of the state after the interference is completed by the second Hadamard gate, consider the action of the n -qubit Hadamard gate on an n -qubit basis state $|x\rangle$

The Deutsch–Jozsa Algorithm

- It is easy to verify that the effect of the **1-qubit Hadamard gate** on a 1-qubit basis state $|x\rangle$ can be written as

$$H|x\rangle = \frac{1}{\sqrt{2}}(|0\rangle + (-1)^x|1\rangle) = \frac{1}{\sqrt{2}} \sum_{z \in \{0,1\}} (-1)^{xz} |z\rangle$$

- Then we can see that the action of the Hadamard transformation on an n -qubit basis state $|x\rangle = |x_1\rangle|x_2\rangle|x_3\rangle\cdots|x_n\rangle$ is given by the action of the n -qubit Hadamard gate on an n -qubit basis state

The Deutsch–Jozsa Algorithm

$$\begin{aligned}
 H^{\otimes n} |x\rangle &= \underbrace{(H \otimes H \otimes \cdots \otimes H)}_n \underbrace{(|x_1\rangle \otimes |x_2\rangle \otimes \cdots \otimes |x_n\rangle)}_n = \underbrace{(H|x_1\rangle) \otimes (H|x_2\rangle) \otimes \cdots \otimes (H|x_n\rangle)}_n \\
 &= \left(\frac{1}{\sqrt{2}}\right)^n \underbrace{\left(|0\rangle + (-1)^{x_1} |1\rangle\right) \otimes \left(|0\rangle + (-1)^{x_2} |1\rangle\right) \otimes \cdots \otimes \left(|0\rangle + (-1)^{x_n} |1\rangle\right)}_n \\
 &= \left(\frac{1}{\sqrt{2}}\right)^n \left(\sum_{z_1 \in \{0,1\}} (-1)^{x_1 z_1} |z_1\rangle \right) \left(\sum_{z_2 \in \{0,1\}} (-1)^{x_2 z_2} |z_2\rangle \right) \cdots \left(\sum_{z_n \in \{0,1\}} (-1)^{x_n z_n} |z_n\rangle \right) \\
 &= \left(\frac{1}{\sqrt{2}}\right)^n \sum_{z_1 z_2 \cdots \in \{0,1\}} (-1)^{x_1 z_1 + x_2 z_2 + \cdots + x_n z_n} |z_1\rangle |z_2\rangle \cdots |z_n\rangle
 \end{aligned}$$

The Deutsch–Jozsa Algorithm

- We are now in a position to derive $|\psi_3\rangle$ by applying the separable transformation $H^{\otimes n} \otimes I$ to $|\psi_2\rangle$
- We only care about the Hadamard part, since it is the output of the data register we will measure. It produces the output

$$\begin{aligned} |\psi_3\rangle &= (H^{\otimes n} \otimes I) |\psi_2\rangle = \left(\frac{1}{\sqrt{2}}\right)^n \sum_{x \in \{0,1\}^n} (-1)^{f(x)} H^{\otimes n} |x\rangle \otimes I \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \\ &= \underbrace{\left(\frac{1}{\sqrt{2}}\right)^n \sum_{x \in \{0,1\}^n} (-1)^{f(x)} H^{\otimes n} |x\rangle}_{\text{We are interested in this part}} \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \end{aligned}$$

We are interested in this part

The Deutsch–Jozsa Algorithm

- By exploiting the previous result

$$H^{\otimes n} |x\rangle = \left(\frac{1}{\sqrt{2}}\right)^n \sum_{z_1 z_2 \dots \in \{0,1\}} (-1)^{x_1 z_1 + x_2 z_2 + \dots + x_n z_n} |z_1\rangle |z_2\rangle \dots |z_n\rangle$$



$$|\psi_3\rangle = \left(\frac{1}{\sqrt{2}}\right)^n \sum_{x \in \{0,1\}^n} (-1)^{f(x)} \overbrace{H^{\otimes n} |x\rangle} \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

$$|\psi_3\rangle = \left[\left(\frac{1}{\sqrt{2}}\right)^n \sum_{x \in \{0,1\}^n} (-1)^{f(x)} \left(\frac{1}{\sqrt{2}}\right)^n \sum_{z_1 z_2 \dots \in \{0,1\}} (-1)^{x_1 z_1 + x_2 z_2 + \dots + x_n z_n} |z_1\rangle |z_2\rangle \dots |z_n\rangle \right] \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

The Deutsch–Jozsa Algorithm

Therefore, the state after the final n-qubit Hadamard gate in the Deutsch-Jozsa algorithm is

$$|\psi_3\rangle = \left(\frac{1}{2}\right)^n \sum_{z_1 z_2 \cdots z_n \in \{0,1\}} \underbrace{\left(\sum_{x_1 x_2 \cdots x_n \in \{0,1\}} (-1)^{f(x_1, x_2, \dots, x_n) + x_1 z_1 + x_2 z_2 + \cdots + x_n z_n} \right)}_{G(z_1, z_2, \dots, z_n)} |z_1\rangle |z_2\rangle \cdots |z_n\rangle \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

where we have regrouped the sum and defined a scalar function $G(z_1, z_2, \dots, z_n)$ of the summation indices $z_1, z_2, \dots, z_n$

The Deutsch–Jozsa Algorithm

So, the final output is an expansion along the z-basis,

$$|\psi_3\rangle = \left(\frac{1}{2}\right)^n \sum_{z_1 z_2 \dots z_n \in \{0,1\}} G(z_1, z_2, \dots, z_n) |z_1\rangle |z_2\rangle \dots |z_n\rangle \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

Therefore

$$\text{\textit{Data register at access point } Q} = \frac{1}{2^n} \sum_{z_1 z_2 \dots z_n \in \{0,1\}} G(z_1, z_2, \dots, z_n) |z_1\rangle |z_2\rangle \dots |z_n\rangle$$

At the end of the algorithm a measurement of the data register is made in the computational basis (just as was done for the Deutsch algorithm)

The Deutsch–Jozsa Algorithm

- To see what happens, consider the total amplitude (coefficient) $G(0)$ of

$$|z\rangle = |0\rangle|0\rangle\cdots|0\rangle \equiv |0\rangle^{\otimes n}$$

- This will tell us something about the other $2^n - 1$ CBS amplitudes, $G(z_1, z_2, \dots, z_n)$; for $z_i > 0, \forall i \in \{1, 2, \dots, n\}$. This amplitude is

$$\frac{G(0, 0, \dots, 0)}{2^n} = \frac{1}{2^n} \sum_{x_1 x_2 \dots x_n \in \{0,1\}^n} (-1)^{f(x_1, x_2, \dots, x_n)}$$

- Consider this amplitude in the two cases: **f constant** and **f balanced**

The Deutsch–Jozsa Algorithm

- If f is constant, the amplitude of $|0\rangle^{\otimes n}$ is either +1 or -1 (depending on what value $f(x)$ takes, i.e. 0 or 1)

$$\frac{G(0,0,\dots,0)}{2^n} = \frac{1}{2^n} \sum_{x_1 x_2 \dots x_n \in \{0,1\}} (-1)^0 = \frac{1}{2^n} \sum_{x_1 x_2 \dots x_n \in \{0,1\}} 1 = \frac{2^n}{2^n} = 1 \quad \text{for} \quad f(x_1, x_2, \dots, x_n) = 0$$

$$\frac{G(0,0,\dots,0)}{2^n} = \frac{1}{2^n} \sum_{x_1 x_2 \dots x_n \in \{0,1\}} (-1)^1 = \frac{1}{2^n} \sum_{x_1 x_2 \dots x_n \in \{0,1\}} (-1) = -\frac{2^n}{2^n} = -1 \quad \text{for} \quad f(x_1, x_2, \dots, x_n) = 1$$

thereby forcing the amplitudes of all other z-basis kets in the expansion to be 0

The Deutsch–Jozsa Algorithm

- Thus, in *the constant* case

Data register at access point Q $= \pm |0\rangle|0\rangle \cdots |0\rangle$

- It turns out that *if f is constant*, a measurement of the data register *is certain to return all 0s* (by ‘all 0s’ we mean the binary string $00 \cdots 0$)

The Deutsch–Jozsa Algorithm

If f is balanced, then the amplitude of $|0\rangle^{\otimes n}$ in the expansion is

$$\frac{G(0,0,\dots,0)}{2^n} = \frac{1}{2^n} \sum_{x_1 x_2 \dots x_n \in \{0,1\}^n} (-1)^{f(x_1, x_2, \dots, x_n)}$$

but a balanced f promises an equal number $f(x_1, x_2, \dots, x_n) = 0$
and $f(x_1, x_2, \dots, x_n) = 1$, so the sum has an equal number of +1s and -1s,
forcing it to be 0

The Deutsch–Jozsa Algorithm

- If Therefore, the probability of a measurement causing a collapse to the state $|0\rangle^{\otimes n}$ is 0 (the amplitude-squared of the CBS state $|0\rangle^{\otimes n}$)
- We are guaranteed to never get a reading of “0” when we measure the data register at access point Q

The Deutsch–Jozsa Algorithm in Summary

- We've explained the purpose of all the components in the circuit and how each plays a role in leveraging *quantum parallelism*, *interference* and *phase kick-back*
- We run the circuit **one time only** and measure the data register output in the standard basis
- If we read “0” then f is constant
- Otherwise f is balanced

Evolution

Evolution

Question

- How does the state, $|\psi\rangle$, of a quantum mechanical system change with time?

Answer

- **Postulate 2:** The evolution of a *closed* quantum system is described by a *unitary transformation*. That is, the state $|\psi\rangle$ of the system at time t_1 is related to the state $|\psi'\rangle$ of the system at time t_2 by a unitary operator U which depends only on the times t_1 and t_2 ,

$$|\psi'\rangle = |\psi(t_2)\rangle = U(t_1, t_2) |\psi(t_1)\rangle$$

Evolution

- Postulate 2 requires that the system being described be closed
- That is, it is not interacting in any way with other systems
- In reality, of course, all systems (except the Universe as a whole) interact at least somewhat with other systems
- Nevertheless, there are interesting systems which can be described to a good approximation as being closed, and which are described by unitary evolution to some good approximation

Evolution

- Postulate 2 describes how the quantum states of a closed quantum system at two different times are related
- A more refined version of this postulate can be given which describes the evolution of a quantum system in ***continuous time***
- From this more refined postulate we will recover Postulate 2

Evolution

- **Postulate 2'**: The time evolution of the state of a closed quantum system is described by the *Schrodinger equation*,

$$i\hbar \frac{d|\psi(t)\rangle}{dt} = H|\psi(t)\rangle$$

- In this equation, $\hbar$ is a physical constant known as *Planck's constant*
- The exact value is not important to us. In practice, it is common to absorb the factor $\hbar$ into H , effectively setting $\hbar = 1$
- H is a fixed Hermitian ($H = H^\dagger$) operator known as the *Hamiltonian* (*Not Hadamard!!*) of the closed system

Evolution

- **Question:** What is the connection between the Hamiltonian picture of dynamics, Postulate 2', and the unitary operator picture, Postulate 2?

$$|\psi(t_2)\rangle = \exp\left[\frac{-iH(t_2 - t_1)}{\hbar}\right]|\psi(t_1)\rangle = U(t_1, t_2)|\psi(t_1)\rangle$$

where we define

$$U(t_1, t_2) = \exp\left[\frac{-iH(t_2 - t_1)}{\hbar}\right]$$

- There is therefore a one-to-one correspondence between the discrete-time description of dynamics using unitary operators, and the continuous time description using Hamiltonians

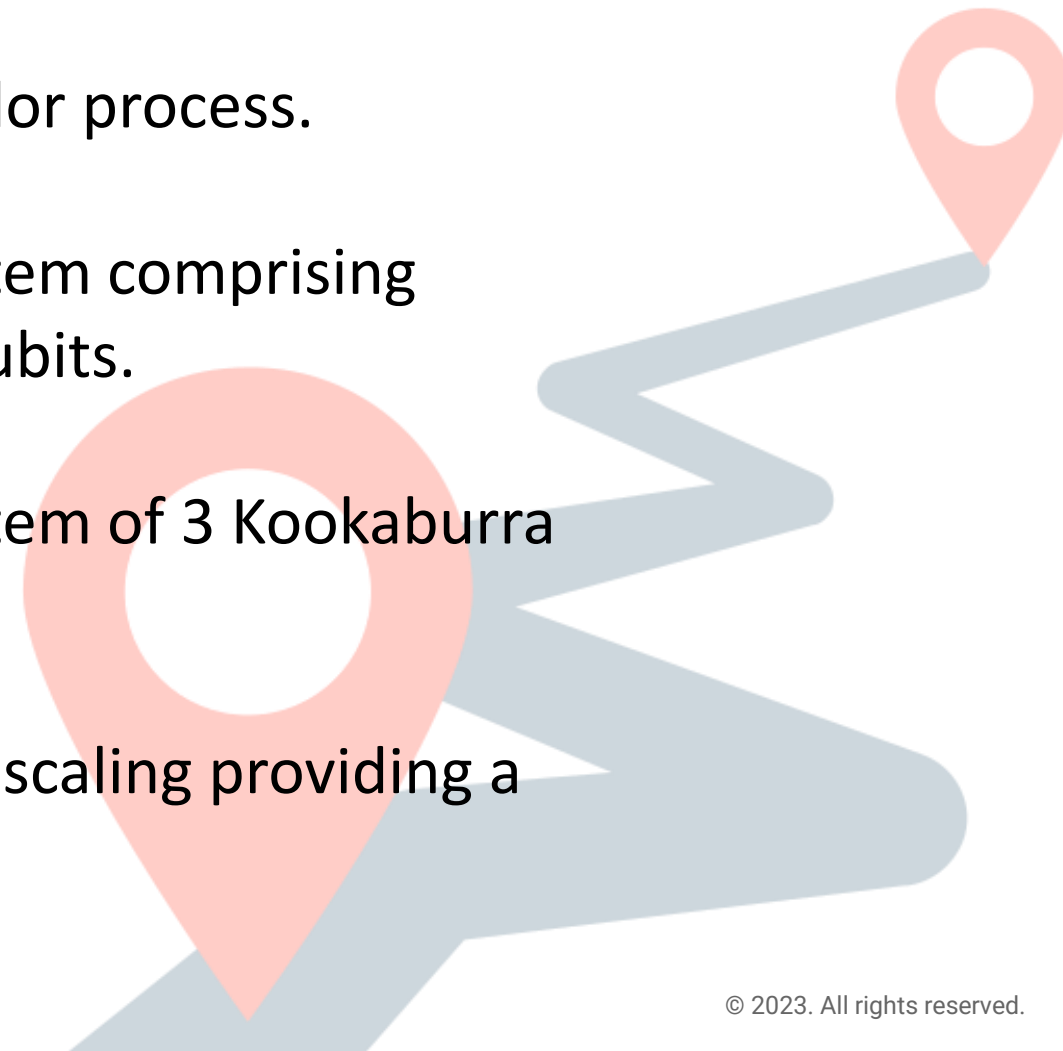
IBM Quantum Development Roadmap

2023: IBM will deliver the 1,121-qubit Condor process.

2024: IBM will demonstrate a quantum system comprising three Flamingo processors totaling 1,386 qubits.

2025: IBM will demonstrate a quantum system of 3 Kookaburra processors totaling 4,158 qubits.

This leap forward will usher in a new era of scaling providing a clear path to 100,000 qubits and beyond.



Google Quantum Development Roadmap

Building a useful quantum computer

Our roadmap for building a useful error-corrected quantum computer with key milestones. We are currently building one logical qubit that we will scale in the future.

